Meteorological Conditions and Rural Injury Accidents in Iceland

Analysis of Accident Frequency and Traffic, 2007–2025

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September 2026

FACULTY OF INDUSTRIAL ENGINEERING, MECHANICAL ENGINEERING AND COMPUTER SCIENCE

M.Sc. thesis
in Software Engineering

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30 ECTS thesis submitted in partial fulfillment of a
Master of Science degree in Software Engineering

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Bibliographic information:
Ásgeir Snær Magnússon (2026). *Meteorological Conditions and Rural Injury Accidents in Iceland*, M.Sc. thesis, Faculty of Industrial Engineering, Mechanical Engineering and Computer Science, University of Iceland.

Reykjavik, Iceland, September 2026

Abstract

This thesis examines rural injury-accident occurrence in Iceland in relation to mean wind, gust, temperature, season, hour, daylight, and available traffic data. Accident records, cleaned 10-minute weather observations, urban boundaries, annual road-section traffic statistics, and daily counter data are combined in a documented analysis. The main analysis includes 6,259 rural injury accidents from 2007–2025 with valid wind observations within 20 km. Within each weather station and study season, observed accidents in each mean-wind, gust, and temperature interval are compared with the number expected from the local background frequency of that interval. Results are shown for all injury accidents and their serious-or-fatal subset, for the complete year and each season. Confidence intervals are obtained from 5,000 bootstrap samples clustered by weather station.

The observed/expected ratio is close to one across most low and moderate wind intervals. At mean wind speeds of 20–25 m/s, 53 accidents were observed compared with 26.4 expected, giving a ratio of 2.01 (95% station-clustered bootstrap interval 1.44–2.63). Gust shows a corresponding upper-tail pattern in the O/E and matched-time analyses. In the matched-time analysis, the odds ratio for mean wind of at least 15 m/s versus 0–5 m/s is 1.61 (95% interval 1.38–1.87), and it changes little when temperature is included. The annual-traffic model also gives an elevated 20–25 m/s rate ratio of 2.27, but its traffic within each wind interval is estimated rather than observed. The stricter daily-counter analysis starts from observed daily traffic and also shows elevated upper-wind estimates, but covers only 762 accidents and must estimate how the daily total is divided among wind intervals. Temperature is available for 6,259 accidents and has a non-monotonic pattern; the matched daylight estimates are imprecise and close to one. The adjusted severity model gives no clear evidence of greater serious-or-fatal severity in strong wind. The result describes relative accident occurrence, not a causal effect or accident risk per vehicle. Complete hourly traffic is unavailable at the required spatial and temporal resolution for the full study period. The thesis documents data cleaning, matching, coverage and statistical analysis separately.

Ágrip

Í þessari ritgerð eru tengsl veðurs og umferðarslysa með meiðslum í dreifbýli á Íslandi rannsökuð fyrir tímabilið 2007–2025. Megináhersla er lögð á vind, en einnig eru skoðuð vindhviður, hitastig, árstíð, tími dags og birta. Slysagögn eru tengd við tíu mínútna veðurmælingar frá nærliggjandi veðurstöðvum og niðurstöðurnar bornar saman við tíðni sömu veðurskilyrða. Umferðargögn eru síðan notuð til að kanna nánar samband vinds og slysatíðni.

Meginniðurstöðurnar byggja á 6.259 slysum sem hægt var að tengja við gilda vindmælingu í innan við 20 km fjarlægð. Fyrir hverja veðurstöð og árstíð var borinn saman raunverulegur fjöldi slysa við þann fjölda sem búast mætti við miðað við hversu oft vindhraði á hverju bili mældist á viðkomandi stað. Við 20–25 m/s meðalvind urðu 53 slys en búast hefði mátt við 26,4. Hlutfall raunverulegs og vænts fjölda slysa var því 2,01 (95% öryggisbil 1,44–2,63). Niðurstöðurnar benda þannig til þess að slys séu hlutfallslega algengari þegar vindur er mikill en tíðni slíkra vindskilyrða ein og sér myndi gefa til kynna. Greiningar þar sem tekið er tillit til umferðar styðja sömu meginniðurstöðu.

Svipað samband kemur fram fyrir miklar vindhviður, en samband hitastigs og slysatíðni er ekki jafn skýrt. Ekki fundust skýrar vísbendingar um að mikill vindur tengdist hærra hlutfalli alvarlegra slysa eða banaslysa meðal þeirra slysa með meiðslum sem þegar höfðu orðið. Niðurstöðurnar sýna tölfræðileg tengsl en sanna ekki að veður valdi slysum. Takmarkanir umferðargagnanna gera jafnframt að verkum að ekki er hægt að reikna nákvæma áhættu á hvert ökutæki fyrir allt rannsóknartímabilið.

Contents

List of Figures

List of Tables

Abbreviations

ADU	Average annual daily traffic.
SDU	Summer daily traffic, June–September.
VDU	Winter daily traffic, December–March.
VHDU	Spring/autumn daily traffic derived from ADU, SDU and VDU.
O/E	Observed accidents divided by expected accidents.

1 Introduction

1.1 General

Weather affects both driving conditions and the decision to travel. Strong wind can make a vehicle harder to control, low temperature can coincide with slippery roads, and darkness can reduce visibility. These conditions are important in Iceland, where weather varies greatly across locations and over short periods.

This thesis studies rural accidents involving injury in Iceland from 2007–2025. Accident times and locations are linked to nearby ten-minute weather observations. The main comparison concerns mean wind speed. Wind gust, temperature, season, time of day, daylight, and the available traffic data are then examined separately.

Accident counts alone are not accident risk. Severe weather can increase the hazard faced by each vehicle while reducing the number of vehicles on the road. Weather frequency and traffic volume are therefore considered separately Andrey et al. (2003); Lord and Mannering (2010); Washington et al. (2020).

1.2 Aim and Research Questions

The aim is to quantify how the occurrence of rural injury accidents varies with weather and what the available traffic data add to the mean-wind result. Three questions guide the analysis:

1. How do observed/expected ratios for all injury accidents and the serious-or-fatal subset vary across mean-wind, wind-gust, and temperature intervals during the year and within each season?
2. Are the weather-frequency results supported when each accident time is compared with matched non-accident times?
3. How does accident occurrence vary with mean wind after including the available annual and daily traffic data, and does the daily-counter result vary by season?

The first question gives the broad 2007–2025 description. The two outcome groups are nested: serious-or-fatal accidents are a subset of all injury accidents. The second question checks the weather result using another time comparison. The third uses smaller linked traffic samples and is interpreted as supporting evidence. The study estimates associations, not causal effects or a complete national accident rate per vehicle-kilometre.

1.3 Thesis Outline

Chapter 2 gives the background for the weather and traffic comparisons. Chapter 3 describes the data, preparation pipeline, and statistical methods. Chapter 4 presents the results, beginning with mean wind. Chapter 5 discusses their interpretation and limitations, and Chapter 6 gives the conclusions. Additional figures and detailed checks are given in the appendix.

2 Background

2.1 Wind and Rural Road Accidents

Crosswind can produce overturning, lateral sideslip, or excessive yaw. The critical condition depends on vehicle geometry and mass, vehicle speed, road friction, wind direction, and driver response rather than wind speed alone Baker (1986, 1994). High-sided vehicles are particularly exposed, but the physical mechanisms also provide a reason to examine loss-of-control accidents more generally. These engineering studies establish physical plausibility and vehicle-specific failure mechanisms; they do not by themselves establish a population-level association between wind and accident occurrence.

This problem is directly relevant in Iceland, where terrain can accelerate and redirect near-surface wind over short distances. An Icelandic reliability study modelled accident probability as the combined outcome of wind velocity and direction, road camber and friction, and vehicle speed Snæbjórnsson et al. (2007). That work explains why a weather station is an imperfect proxy for conditions at a specific road location and why results should be compared under several distance limits. Mobile road- measurement research similarly shows that roadside terrain creates marked spatial variation that a sparse fixed-station network cannot fully resolve Pu et al. (2018).

Vegagerðin has developed a vehicle-wind model to support decisions about hazardous conditions and road closures, while also noting the limited experimental evidence available for passenger cars Hrafnkelsdóttir (2020). The present thesis does not test that operational model; it examines population-level accident occurrence using a separate data design.

Empirical work has compared the share of accidents occurring in high wind with the share of time that high wind was observed, an approach closely related to the observed/expected comparison used here Edwards (1994). Studies linking crashes to nearby weather stations likewise show that station observations can be informative while retaining measurement error Young and Liesman (2007). More broadly, reviews find that road-safety evidence for wind is less consistent than for precipitation Theofilatos and Yannis (2014).

2.2 Temperature, Daylight, and Time of Day

Temperature, daylight, and time of day add context that wind alone cannot provide. They affect road-surface conditions, visibility, travel patterns, and the composition of road users, but accident counts in these categories are not rates unless the corresponding amount of travel is known. Temperature effects can also be non-linear and season-specific: a Seoul time-series study found its strongest cold-weather association below freezing in winter, while an hourly case-crossover study in Shenzhen found a J-shaped temperature relationship Lee et al. (2014);

Zhan et al. (2020). Temperature is therefore compared with its local seasonal frequency rather than treated as having one linear effect.

Darkness presents the same denominator problem. A method developed using Norwegian, Swedish, and Dutch data compared daylight and darkness within clock time and season, avoiding the interpretation of raw night-time counts as risk Johansson et al. (2009). Here, raw hour, season, and daylight counts describe the sample. Matched daylight and adjusted severity models are used for numerical comparisons.

2.3 Traffic and Accident Rates

Weather can change the amount of traffic as well as the hazard. Research using automatic traffic recorders shows that extreme weather variables, including wind, can reduce or redistribute hourly traffic volume Sathiaraj et al. (2018). Accident counts during strong wind can remain low even if risk to each travelling vehicle increases. High-resolution crash studies show the value of hourly weather and matched-time analyses rather than daily or monthly averages Xing et al. (2019). This thesis reports three distinct quantities: accident occurrence relative to weather frequency, rates based on estimated vehicle-kilometres, and changes in observed daily traffic. Studies with real-time traffic, weather, and road geometry show what a more complete analysis could achieve when such data are available Ahmed et al. (2012).

3 Data and Methods

3.1 Data

3.1.1 Source Data

Table 3.1 lists the downloaded source data. The identifiers in the first column are used in Table 3.3. The source documentation is cited in the bibliography Samgöngustofa (2026); Veðurstofa Íslands (2026); Vegagerðin (2026); Hagstofa Íslands (2026).

Table 3.1: Downloaded source data. The study period is 2007–2025.

ID	Source website	Downloaded data	Records and period
ACC	Samgöngustofa	Accident events and road links	126,607 events and 126,610 road links, 2007–2025
IMO	Icelandic Meteorological Office	Ten-minute weather observations and station metadata	232,459,562 observations from 338 stations; 438 station records, 2007–2025
VGD-A, VGD-D, VGD-R	Vegagerðin	Annual road traffic, daily counter reports, counter locations, and road geometry	33,757 section-years, 2000–2025; 1,033,659 counter channel-days, 2019–2024
SI	Statistics Iceland	Urban-area polygons	97 polygons, 2020–2024 publication

The 2025 accident, injury, vehicle, road-link, annual-traffic, and weather records pass through the same preparation rules as the earlier years and are included in the analysis. Daily counter reports currently end in 2024.

Table 3.2 shows the first chronological records in the canonical rural injury-accident file. The source identifier is named `nid` or `NID` in the supplied files and is renamed `id` at import. Its values are source record keys, not row numbers, so chronological records are not expected to be numbered 1, 2, 3, and so forth. They are kept unchanged because they link the accident, road-link, and vehicle files. The original injury and accident-type codes are also kept. The table is generated from `data/analysis/accidents.csv`; it is not typed into the thesis.

Table 3.2: *First chronological records in the canonical rural injury-accident analysis file*

id	Date	Time	Lat.	Lon.	meidsli	tegohapps	Road section
3167342	2007-01-02	11:18	64.0523	-21.5325	3	12	000-102
3167947	2007-01-03	03:30	64.9063	-13.9619	3	12	96-07
3167977	2007-01-03	13:00	65.9231	-23.4399	3	12	60-44
3168112	2007-01-03	18:04	65.5153	-18.6033	3	240	1-p1
3168111	2007-01-03	18:08	64.5771	-21.8793	3	12	1-g8
3168229	2007-01-03	23:37	64.0538	-21.4916	3	11	1-d9
3168293	2007-01-04	02:22	64.0233	-22.6323	3	11	429-02
3168875	2007-01-05	06:24	64.3115	-20.2533	3	910	35-08
3168970	2007-01-05	10:41	64.7480	-14.4489	3	12	1-u5
3169241	2007-01-05	17:23	64.0188	-21.3859	3	240	1-d8

Table 3.3 records the exact file transformations used to create the analysis data. Dataset contents and purposes have already been defined in Table 3.1 and the preceding text. To keep the table compact, script paths are relative to `src/` and data paths are relative to `data/`. Collections of unchanged annual workbooks and daily reports are shown with a file pattern. Within a cell, each shared directory is printed once and the following filenames belong to that directory.

Table 3.4 summarises the complete research workflow at the level of analytical stages. After the preparation stages, every analysis uses compact CSV files rather than files in `raw/` or `processed/`. The complete script-by-script inventory, including exact paths, helper steps, and descriptive outputs, is retained in `docs/pipeline.md`.

Table 3.5: *Definitions used throughout the thesis*

Term	Definition	Use in this thesis
`f`	Ten-minute mean wind speed, m/s	Main weather measure, matched within five minutes of the accident time.
`fg`	Wind gust reported in the same ten-minute observation, m/s	Additional weather measure; not a daily maximum.
`t`	Ten-minute air temperature, °C	Additional weather measure matched independently.
O/E	Observed accidents divided by accidents expected from local weather frequency	Main relative-occurrence result; traffic is not included.
ADU	Average vehicles per day over the year	Used to check and derive annual traffic quantities.
SDU	Average vehicles per day in June–September	Official summer traffic value.
VDU	Average vehicles per day in December–March	Official winter traffic value.
VHDU	Average vehicles per day in April–May and October–November	Derived from ADU after SDU and VDU totals are removed.

Table 3.3: *Exact inputs and outputs of the data-preparation pipeline*

Script	Input file(s)	Output file(s)	Description
accidents/build.py	*raw/accidents/* accidents_*.txt vehicles_*.txt road_links_2007_2025.csv urban_boundaries_2020_2024.geojson	*processed/accidents/* all.csv	Joins accident, vehicle, road-link, and boundary data.
weather/download_weather.py	*officialsuppliedURL/* stod.txt, f_*.txt, fj_*.txt, fv_*.txt	*raw/weather/* stations.csv weather_10min_raw.parquet weather_10min_raw_audit.csv	Downloads the complete official station-file delivery and combines study-period rows without filtering.
weather/clean.py	*raw/weather/* weather_10min_raw.parquet	*processed/weather/* weather.parquet cleaning.csv	Applies the fixed weather-quality rules and records annual counts.
weather/frequency.py	*processed/weather/* weather.parquet	*processed/weather/* frequency.csv yearly_frequency.csv temperature_frequency.csv traffic_frequency.csv	Counts pooled and yearly local weather frequency for O/E and traffic models.
traffic/annual.py	*raw/traffic/annual/* *.xls*	*processed/traffic/* annual.csv	Standardises road section, length, ADU, SDU and VDU.
accidents/match_weather.py	*processed/accidents/* all.csv *processed/weather/* weather.parquet *raw/weather/* stations.csv	*processed/accidents/* rural_injury.csv	Matches wind and temperature independently within the stated time and distance limits.
accidents/case_control.py	*processed/accidents/* rural_injury.csv *processed/weather/* weather.parquet	*processed/accidents/* case_control.csv	Selects matched non-accident weather times from the same clean weather source.
traffic/build_road_period.py	*processed/traffic/* annual.csv *processed/accidents/* rural_injury.csv *processed/weather/* weather.parquet *raw/weather/* stations.csv *raw/traffic/reference/* road_section_midpoints.csv, road_sections.parquet	*processed/* weather/road_period_frequency.csv traffic/road_period.csv	Builds road-period mean-wind and traffic rows.
traffic/rate_weather.py	*processed/* traffic/road_period.csv accidents/rural_injury.csv weather/weather.parquet *raw/weather/* stations.csv	*processed/accidents/* rate.csv	Aligns accident wind and temperature with the road-exposure station.
traffic/daily.py	*raw/traffic/daily_pdf/* *.pdf	*processed/traffic/* daily_raw.csv	Parses one daily count per counter channel and date.
traffic/download_roads.py	VGD-R MapServer layer 6	*raw/traffic/reference/* roads.geojson	Downloads the unchanged public road reference.
traffic/locate_counters.py	*processed/traffic/* daily_raw.csv *raw/traffic/reference/* roads.geojson	*processed/traffic/* daily.csv	Combines directional channels and locates counters.
traffic/daily_weather.py	*processed/* traffic/daily.csv weather/weather.parquet *raw/weather/* stations.csv	*processed/traffic/* daily_match.parquet daily_weather.csv	Matches counter-days to a nearby weather station.
traffic/accident_wind.py	*processed/* accidents/rural_injury.csv traffic/daily_weather.csv traffic/locations.csv weather/weather.parquet *raw/weather/* stations.csv	*processed/traffic/* accident_wind.csv	Matches accidents to the assigned counter-day station.
export_tables.py	*processed/* accidents/rural_injury.csv accidents/rate.csv accidents/case_control.csv weather/frequency.csv weather/yearly_frequency.csv weather/traffic_frequency.csv weather/cleaning.csv traffic/annual.csv traffic/road_period.csv traffic/daily_weather.csv traffic/accident_wind.csv traffic/locations.csv	*analysis/* analysis CSV files listed below	Selects only the variables used by ordinary analysis.

3.1.2 Accident Data

The accident source covers 2007–2025 Samgöngustofa (2026). The study population comprises 6,414 rural accidents involving injury. Accident severity is based on the source field `meidsli`: code 1 denotes a fatal accident, code 2 a serious accident, code 3 a minor-injury accident, and code 4 an accident without injury. The main outcome is `meidsli` ≤ 3. The serious-or-fatal comparison uses `meidsli` ≤ 2. The source data also contain 17,591 rural damage-only accidents; these are outside the injury outcome.

Rural status is derived from accident coordinates and Statistics Iceland urban-locality polygons Hagstofa Íslands (2026). Points inside an urban polygon are classified as Urban, and valid points outside all polygons as Rural. A defined Vestmannaeyjar area is also classified as Urban. The same 2020–2024 polygon dataset is applied to the full 2007–2025 accident period. Changes in urban boundaries before 2020 are not represented.

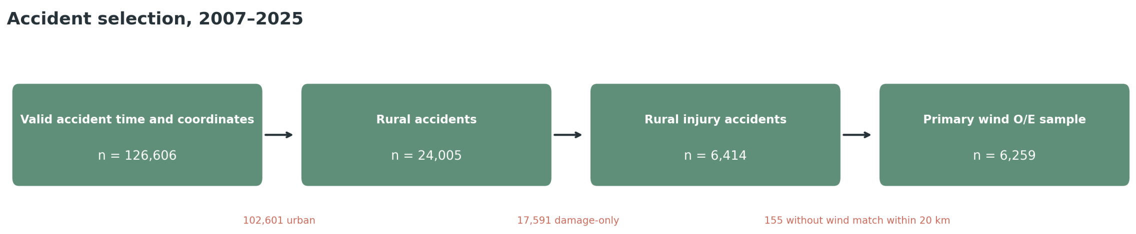

Figure 3.1: Available accident records, severity of the rural injury sample, and coverage of the main weather match. Damage-only accidents remain available but are not part of the injury outcome.

3.1.3 Weather Data

The weather source contains 10-minute observations from Icelandic weather stations. The variables used here are ten-minute mean wind speed (`f`), reported wind gust (`fg`), and temperature (`t`). For each accident, the gust is taken from the same matched ten-minute observation as mean wind; it is not the maximum across the accident day. Mean wind speed remains the main weather measure. A wind observation is used when

$$0 \leq f < 45, \qquad 0 \leq fg < 65, \qquad fg + 0.5 \geq f.$$

Rows missing either `f` or `fg` are excluded. Negative values, values at or above either upper limit, and internally inconsistent values where `fg = 0` but `f > 0` are excluded and reported in the annual audit. A value of `f = 0` alone is kept. Only uninterrupted all-zero runs where `f = fg = 0` for at least two hours are excluded as potentially frozen sensors. Missing temperature does not affect whether a wind observation is used.

Each accident is matched to the geographically nearest station with a clean wind observation at either adjacent timestamp on the 10-minute observation grid. The observations have 10-minute resolution, but the selected observation is no more than five minutes from the recorded

Table 3.4: *Main analysis pipeline and reproducible data products*

Analysis stage	Main input	Main output	Purpose
Accident data preparation	Source accident, road-link and urban-area files	`accidents.csv`	Selects rural injury accidents and keeps traceable source IDs.
Weather preparation and matching	Ten-minute weather and station files `accidents.csv`	`weather_frequency.csv` `accident_conditions.csv` `case_control.csv`	Cleans weather and links accident and control times.
Weather-frequency analysis	`accident_conditions.csv` `weather_frequency.csv`	`oe_results.csv` `weather_oe_panels.csv` three five-panel figures	Compares accident shares with local weather-time shares.
Traffic data preparation	Annual road traffic, road geometry and daily counters	`road_rate.csv` `daily_traffic.csv` `counter_wind.csv`	Makes linked annual and daily traffic inputs.
Traffic-adjusted wind analysis	Prepared traffic inputs `accidents.csv`	annual rate results `daily_season_panel.csv` daily and seasonal estimates	Estimates rates using annual traffic and allocated daily vehicles.
Supporting analyses	`case_control.csv` weather and traffic analysis files	matched-time, daylight, severity and traffic-response results	Checks and helps interpret the main comparisons.
Validation and thesis products	Analysis CSVs `docs/pipeline.md`	validation report thesis tables and figures	Checks numbers and builds reported products.

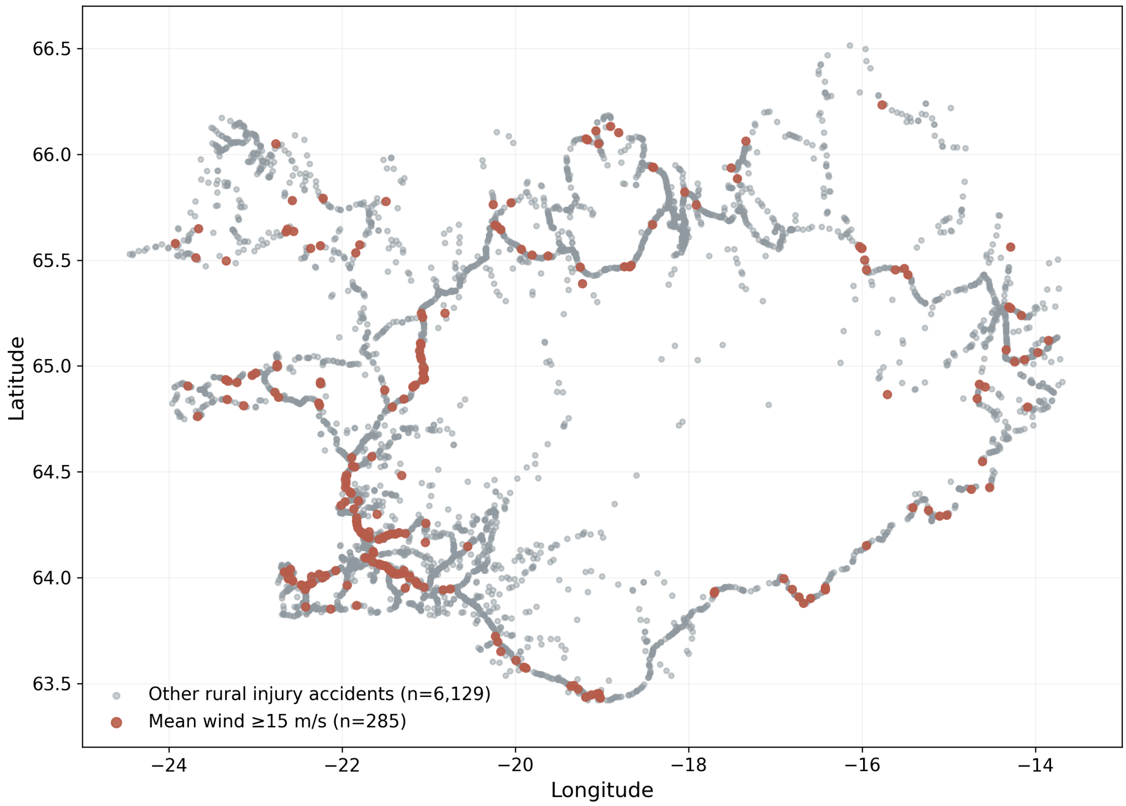

Figure 3.2: Locations of the 6,414 rural injury accidents. The 300 accidents matched to mean wind of at least 15 m/s are highlighted. The coordinates show the spatial coverage of the study; the figure is not a regional rate map.

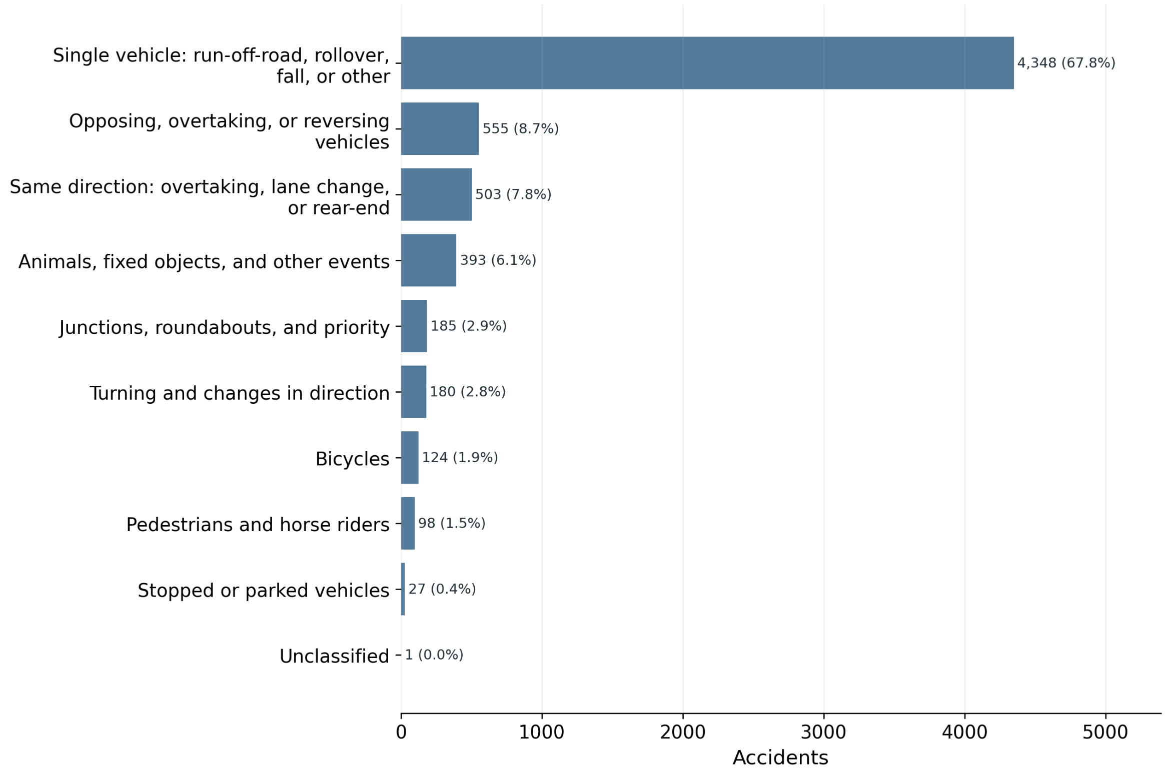

Figure 3.3: Rural injury accidents by accident type.

accident time. The main maximum distance is 20 km. The matching file includes all 6,414 accidents and records whether each accident meets the weather-match criterion.

Temperature is matched independently because the nearest station with valid wind does not always report temperature. The nearest temperature observation between -30 and 30°C within the same 20 km and five-minute limits is selected. The temperature station, distance, and time difference are stored separately. Wind and temperature may come from different stations. This increases usable temperature coverage without changing the wind sample or discarding accidents with missing temperature.

The Icelandic Meteorological Office supplied the observations as separate station files with five fields: time, station, temperature, mean wind, and gust. The delivery contains 214 current observation files, 124 older complementary files, and 123 Road Administration station files. The older files contain no observations in 2007–2025, but are retained in the raw source audit. All study-period records are combined once, then the same clean weather file is used for accident matching, background frequencies, matched control times, and traffic analyses. No temperature value is imputed.

Table 3.6 reports spatial and temporal match quality for wind and temperature. It shows both matched and unmatched accidents, making the greater loss of temperature observations visible before the later analyses.

Table 3.6: Share of accidents with weather information from a station with a valid measurement within 20 km. Distance is shown as median / P90; every retained observation is within five minutes of the accident time.

Variable	Accidents	Share	Stations	Distance (km)
Mean wind and gust	6,259	97.6%	279	5.1 / 12.3
Temperature	6,259	97.6%	279	5.1 / 12.3

Table 3.7: Wind-data scope and quality audit, 2007–2025. Quality-rule shares use the rows in station-years containing wind data as their denominator. Every delivered station-year contains at least one wind measurement.

Category	Records	Share
All delivered station-time rows	232,459,562	100.00%
Rows in station-years containing wind data	232,459,562	100.00%
Missing `f` or `fg`	261,519	0.11%
Negative or upper-threshold wind	8,766	0.00%
Internally inconsistent `f`/`fg`	257,245	0.11%
Frozen all-zero runs ($\geq$ 2 hours)	1,473,082	0.63%
Clean wind observations retained	230,458,950	99.14%

The delivery contains 232,459,562 study-period records. The fixed wind-quality rules exclude 2,000,612 records and retain 230,458,950 (99.14%). The largest excluded category is uninterrupted all-zero runs, which is stated separately because it reflects the rule for possible instrument stoppages rather than a missing-value code. The detailed audit reports each rule separately. The audit and the weather-distribution figure use the full 2007–2025 weather delivery. For each accident's station and season, the O/E expected count uses all clean background observations in that station–season group, not only observations recorded on accident days.

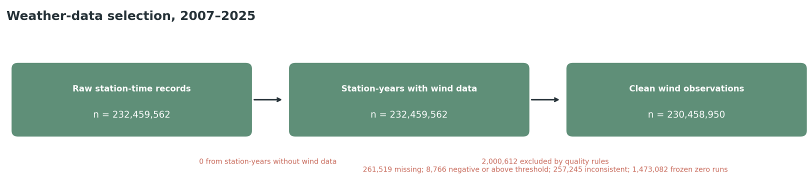

Figure 3.4: Weather observations retained after the documented wind-quality rules.

3.1.4 Traffic Data

Two traffic sources are used for different purposes.

Annual road-section files provide average annual daily traffic (ADU), summer daily traffic (SDU), winter daily traffic (VDU), section length, and annual vehicle-kilometres for 2000–2025 Vegagerðin (2026). The road-section comparison uses the years shared with the accident study, 2007–2025. Following the official definitions, SDU covers June–September and VDU covers December–March. These files can be linked to accidents with a registered road-section code, but exact matching is much narrower than the weather matching.

Table 3.8: Official traffic averages and the derived spring/autumn value used in the annual-traffic analysis.

Period	Months	Daily traffic reference
ADU (*ársdagsumferð*)	Full calendar year	Official annual daily traffic
VDU (*vetrardagsumferð*)	December–March	Official winter daily traffic
SDU (*sumardagsumferð*)	June–September	Official summer daily traffic
VHDU (derived)	April–May and October–November	Residual daily average calculated from ADU, SDU, VDU, and the number of days in each period

Throughout the analyses, the four study seasons are aligned with these traffic periods: winter is December–March, spring is April–May, summer is June–September, and autumn is October–November. These are study-specific traffic-aligned seasons rather than conventional three-month meteorological seasons. VHDU is not calculated as $ADU - SDU - VDU$. Annual traffic implied by ADU is first converted to a calendar-year total; the day-weighted SDU and VDU totals are subtracted, and the remainder is divided by the number of spring and autumn days:

$$VHDU = \frac{D_{year}ADU - D_{VDU}VDU - D_{SDU}SDU}{D_{VHDU}}.$$

Here, each D is the number of calendar days in that period. VHDU is therefore a residual estimate rather than an additional official traffic measure.

Daily traffic counts extracted from Vegagerðin reports cover 2019–2024. They describe observed traffic at counter sites and are useful for measuring travel demand during windy days. They do not provide hourly vehicle-kilometres at every accident location. More than one counter may also occur on a road section. Each report identifies a counter by road section and *stöð* (road station). The station value is placed along the official road geometry and marked as an estimated location; annual-file start and end stations are used only as a fallback where the official station range is unavailable. They are therefore used only to describe whether travel demand changes on windy days; they are not part of the main O/E calculation.

Of 68,946 annual road-section–year–period records, 60,595 have nearby usable wind data; separately, 738,424 of 774,274 daily counter-days have a matched daytime wind summary. The more restrictive accident samples used by the traffic models are reported in Chapter 4.

3.2 Data Preparation and Reproducibility

The repository groups the code into weather, accident, traffic, table, and figure modules. Preparation scripts clean and match source records, table scripts calculate numerical results, and figure scripts draw completed tables. The repository README identifies the individual programs. After the authorised source deliveries have been placed locally, the reported results can be regenerated with

```
.venv/bin/python -m src.prepare --stage prepare
.venv/bin/python -m src.analyze
```

Figures are generated by the figure scripts and are not manually edited.

3.3 Analysis

3.3.1 Weather-Frequency Comparison

Accident-time mean wind speed `f` is the main weather measure and is divided into 5 m/s intervals. It is the ten-minute mean wind observation matched within five minutes of the accident time, not an instantaneous measurement. Wind gust `fg` from the same observation and independently matched temperature are additional weather measures. For each measure, an accident is compared with the distribution at its matched station in the same study season, pooled across all study years. Expected counts are based on how often each condition occurred locally rather than on a national weather distribution. For example, if 5% of the relevant station–season observations fall in one wind interval, approximately 5% of that group's accidents would be expected there under weather-frequency-proportional occurrence.

Temperature uses the same station-and-season O/E structure with independently matched temperature stations. The intervals are below -6, -6 to -3, -3 to 0, 0 to 3, 3 to 6, 6 to 9, 9 to 12, 12 to 15, and at least 15°C. The seven closed intervals have equal 3°C widths; the two open tails retain every valid match.

Let g identify a weather station and study season. Let A_{jg} be the number of accidents in weather interval j and group g, and let $A_g = \sum_j A_{jg}$. From all clean 10-minute observations, let T_{jg} be the number in interval j and T_g the total in group g. The local background frequency is

$$p_{jg} = \frac{T_{jg}}{T_g}.$$

If accident timing followed local wind frequency, the expected number in the interval would be $A_g p_{jg}$. The pooled observed/expected ratio is

$$R_j = \frac{\sum_g A_{jg}}{\sum_g A_g p_{jg}}.$$

A value of one means that the accident share equals the local time share of the weather interval. A value above one means that accidents are over-represented in that interval. Standardisation within station and season prevents locations or seasons with different weather distributions from dominating the result.

The same comparison can be written as the probability identity

$$\frac{P(A \mid W_j)}{P(A)} = \frac{P(W_j \mid A)}{P(W_j)},$$

where A denotes an accident and W_j weather interval j. The O/E ratio estimates the right-hand side: the share of accidents in an interval divided by that interval's local share of weather time. This follows Bayes' rule as an identity; it is not a Bayesian model and uses no prior or posterior distribution.

Uncertainty is estimated with 5,000 weather-station-clustered bootstrap replicates. Stations are sampled with replacement, while all seasons, weather observations, and accidents associated with a sampled station remain together. The reported interval is the 2.5th to 97.5th percentile of the bootstrap distribution.

3.3.2 Matched-Time Comparison

This is a time-stratified case-crossover design: each accident forms one matched set. The accident time is the case time, and the control times are the other occurrences of the same weekday and clock time in the same calendar month and year. For example, a Tuesday accident at 14:20 is compared with the other Tuesdays at 14:20 in that month. Mean-wind and gust controls use the accident's matched wind station; temperature controls use its independently matched temperature station. Control weather must satisfy the same five-minute observation-time rule, and both station matches are restricted to 20 km. Conditional logistic regression compares weather at the accident time with weather at its matched control times. This time-stratified referent strategy was selected to avoid overlap bias and to control calendar time without selecting controls from outside the accident month Janes et al. (2005).

The model therefore asks whether weather at the accident time differs from weather at otherwise comparable non-accident times. This design controls exactly for weather station, year, month, weekday, and clock time. It controls recurring travel schedules more tightly than the seasonal O/E analysis without estimating hourly traffic from daily counts. It does not control unusual traffic changes, road conditions, or other time-varying causes on a particular day. Continuous effects are reported per 5 m/s or 5°C, together with categorical estimates. The pre-specified mean-wind O/E remains the main result.

Mean wind and temperature are also entered together as categorical variables in one matched-time model. This checks whether the wind estimate persists when temperature at the same case and control times is included. Daylight is calculated at the accident location for the accident date and the same matched dates. Because month, weekday, and clock time are held fixed, only strata that change between darkness, civil twilight, and daylight contribute information.

Seasonal differences in the mean-wind association are tested in a separate matched-time model. Mean wind is grouped as 0–10, 10–15, and at least 15 m/s for stability. The model includes interactions between wind interval and season, and a likelihood-ratio test compares it with a model having one common wind association across seasons. The omnibus test is used instead of judging seasonal differences from separate confidence intervals.

A separate logistic model describes serious-or-fatal rather than minor injury among accidents with complete wind and temperature matches. It includes broad categories of mean wind, temperature, daylight, time of day, and season. The result concerns injury severity among recorded accidents. It is not a model of whether an accident occurs and does not replace the traffic analyses.

3.3.3 Use of Traffic Data

The main analysis uses background wind time. Traffic is not included because it is unavailable at the spatial and temporal resolution required for the full study period. The descriptive estimated-rate calculation uses annual traffic on every eligible road section, including sections without an accident. The main conditional model has a narrower purpose and sample: it includes 4,933 accidents in road-section–year–traffic-period strata that contain an accident and have a valid weather and traffic link. Both calculations use SDU and VDU where they are published, and a clearly labelled residual VHDU estimate for April–May and October–November.

Using those traffic data in the main curve would change the study population and mix seasonal average traffic with 10-minute accident-time wind. The curve measures *wind-frequency-adjusted relative accident occurrence*. It does not itself measure accident rate per vehicle.

3.3.4 Road-Section Traffic Comparison

Traffic is included where the spatial and temporal keys can be linked explicitly. For road section, year, official traffic period, and wind interval k, j, estimated vehicle-kilometres are

$$V_{kj} = q_k L_k D_k p_{kj},$$

where q_k is SDU, VDU, or derived VHDU vehicles per day, L_k is section length, D_k is the number of calendar days, and p_{kj} is the local share of clean 10-minute observations in the wind interval. The accident rate is

$$\text{Rate}_j = \frac{A_j}{\sum_k V_{kj}} \times 100{,}000{,}000.$$

This method accounts for road length and for spatial and seasonal differences in published traffic. It still assumes that traffic within a traffic period is distributed across wind intervals in proportion to time. It does not observe traffic at the accident minute. The station nearest the road-section midpoint with qualifying data supplies p_{kj}. Each accident is then matched again, specifically to that road-section station; the station used by the main O/E match is not carried into this model. An accident is included only when the road-section station is within 20 km and has a valid observation within five minutes. Thus the `weather_station_id` in the compact model input is the same station for the accident count and its estimated denominator.

The main traffic result is a conditional Poisson model. Each road section, year, and traffic period has its own intercept. For stratum k and wind interval j, the model is

$$\log E(A_{kj}) = \alpha_k + \beta_j + \log V_{kj},$$

where A_{kj} is the accident count and $\log V_{kj}$ is the offset. The exponentiated coefficient $\exp(\beta_j)$ is the estimated rate ratio relative to 0–5 m/s within otherwise identical road-section–year–period strata.

The model is fitted separately for all injury accidents and for serious or fatal accidents. A seasonal extension uses year-specific station frequencies for winter, spring, summer, and autumn. Because individual 5 m/s upper bins are sparse within seasons, that extension uses 0–10, 10–15, and at least 15 m/s. Winter and summer use published VDU and SDU, respectively; spring and autumn use the same derived VHDU daily volume but retain separate weather frequencies and accident counts.

Two additional models use the same road, year, traffic-period, and station alignment. The first replaces wind frequency with temperature frequency and uses the nine fixed temperature intervals, with 0 to 3°C as the reference. A traffic period is used only when temperature observations cover its complete season definition; both spring and autumn must be present for VHDU. Accident temperature is measured at the road-assigned station rather than the independently selected temperature station used in the O/E analysis. The second version fits one-vehicle and multiple-vehicle accident counts separately, using the coarse wind intervals 0–10, 10–15, and at least 15 m/s to avoid sparse subgroup cells. Both use estimated exposure and do not add minute-level traffic observations.

The descriptive rate and conditional model include different road records. The descriptive rate sums estimated vehicle-kilometres over every eligible road section. The conditional model includes only road-section–year–period strata containing at least one accident, because all-zero strata provide no within-stratum information: after conditioning on a stratum's total accident count, a total of zero contributes no likelihood for comparing its wind intervals. Those roads remain in the separate descriptive rate that sums all eligible exposure. Wind intervals with positive estimated vehicle-kilometres are kept within those groups. The two traffic results answer different questions and are not expected to reproduce the same rate ratios.

The annual source contains 22,982 road-section/year rows for 2007–2025. There are 1,509 rows with nonpositive published VDU and 552 rows with a nonpositive derived VHDU residual. The affected period is excluded rather than imputed. Seasonal patterns such as SDU below VDU are audited but are not automatically treated as errors.

The daily counter analysis is kept separate from the vehicle-kilometre model. It describes whether traffic at selected counters is lower on windier days; it does not directly observe vehicle-kilometres in each 10-minute wind interval. An illustrative direction check rescales the annual-model denominator using the observed/expected daily traffic percentage in the corresponding full-day mean-wind interval. This deliberately combines unlike resolutions and periods, so it is not a corrected rate estimate. It shows only what direction the annual rate ratios would move if the 2019–2024 daily pattern at selected counters also applied within the annual traffic periods.

3.3.5 Daily Traffic Comparison

Daily traffic is analysed separately because its weather definition is daily. Directional channels are first summed to one 24-hour count for each physical counter and date. The unit is then one fixed counter on one date; separate counter sites on the same road section are not summed. For each counter, year, month, and weekday, expected traffic is the arithmetic mean of the available daily counts in that same group. The traffic index is

$$I_{cd} = 100 \times \frac{Q_{cd}}{\mathrm{mean}(Q_{c,\text{same year, month, weekday}})}.$$

An index below 100 indicates fewer counted vehicles than the mean for that local calendar context. For each wind interval, the observed-to-expected traffic ratio is aggregated across counter-days; 100 denotes the calendar-standardised expected traffic. The arithmetic mean is used so expected and observed totals balance within a complete calendar stratum; medians are used only in separate counter-quality summaries. It describes the participating counter sites rather than national traffic. The calculation is also repeated within VDU, SDU, and VHDU to show whether the traffic response is confined to one part of the year. The 24-hour traffic count is not converted into hourly traffic. Mean wind is calculated from clean observations between 10:00 and 21:59, so this is a daily association rather than an estimate of traffic in that window. The source contains 5,432 counter-days with a reported count of zero. They are kept because the reports do not distinguish a genuine zero or road closure from counter downtime. Consequently, the daily result is supplementary and the result with zero counts excluded should be inspected before interpretation. The report's *stöð* is placed proportionally between the official start and end stations on the registered road-section geometry. The resulting coordinate is explicitly marked as estimated rather than treated as a surveyed counter position. This locates 2,325 of 2,332 counter-site years; the seven unresolved site-years remain without a weather match. An independent check finds an official 20 m road-station point within 10 station metres for 435 of 473 distinct located sites; among those matches, the median coordinate difference is 5.0 m and the 90th percentile is 9.0 m. Failure to find a point within that narrow query window is recorded as unresolved rather than silently accepted. For each located counter, the nearest station within 20 km having usable observations that day is selected. Uncertainty in the daily traffic estimate is obtained by resampling whole counters 1,000 times.

Counter-year mean traffic is compared with exact road-section ADU as a consistency check. A near-complete counter-year has at least 300 observed days; single-counter road sections are evaluated separately because multiple PDF records can represent different locations, directions, or lanes.

Sustained wind is described by the number of valid ten-minute observations in each counter-day with mean wind of at least 15 m/s; six observations equal one hour. Days must contain at least 108 valid observations. Traffic is compared across 0, more than 0–2, 2–6, and at least 6 hours of strong wind using the same counter–calendar standardisation.

The allocated daily-counter rate model uses a stricter accident linkage. Each 2019–2024 accident requires a counter on the exact registered road section and year within 20 km, positive traffic on its date, and qualifying accident-time mean wind. For counter (c), date (d), and wind interval (j), the observed 24-hour count is allocated according to the counter station's valid ten-minute observations:

$$\widehat{Q}_{cdj} = Q_{cd}\frac{N_{cdj}}{N_{cd}}.$$

Accidents are classified by accident-time `f` from the same station used for the counter-day denominator. That station must be within 20 km of both the accident and counter, and the accident observation must be within five minutes of the recorded time. Estimated traffic and accidents are aggregated by counter, year, and wind interval, and a conditional Poisson model gives each counter–year its own intercept. The allocation assumes uniform traffic across valid observation times; it is an estimated within-day traffic split, not observed hourly traffic. The full-day-mean model is shown only as a day-level check in the appendix. The main allocation uses all sufficiently complete 24-hour weather records. A separate check assumes all daily traffic occurs from 07:00 to midnight, uses only weather observations and accidents in that interval, and is reported only to show the effect of that unverified assumption. Both versions are also fitted for the serious-or-fatal outcome where counts permit.

For the seasonal traffic-standardised O/E description, the same daily-counter data are grouped by counter, year, and season. Within every group containing an accident, the observed accident total is redistributed across the three wind intervals in proportion to the allocated vehicle exposure. Expected counts therefore sum exactly to observed accidents within each group. Results are then pooled by season, and uncertainty is obtained from 5,000 bootstrap samples of whole counters. This is a descriptive standardisation within accident-containing groups; the conditional Poisson model remains the inferential traffic comparison. A likelihood-ratio test compares models with and without all six wind-by-season interaction terms. A narrower secondary test compares models with and without the three at-least-15 m/s-by-season terms while retaining the 10–15 m/s interactions in both models. Reporting both tests avoids selecting only the more favourable seasonal comparison.

The seasonal daily-counter analysis uses 0–10, 10–15, and at least 15 m/s. A programmed audit also constructed the requested 5 m/s intervals. In the 20–25 and at-least-25 m/s cells it found only 2 and 1 winter accidents, 1 and 3 spring accidents, 1 and 1 summer accidents, and 2 and 0 autumn accidents. Those cells are too sparse for a useful seasonal comparison, so the coarser intervals are retained for the traffic analysis. The broad weather-frequency analysis has substantially more data and retains its original finer intervals.

For the combined wind O/E figure, two traffic-standardised summaries are calculated beside the primary weather-frequency result. In the annual summary, each road-section–year–traffic-period accident total is distributed according to estimated vehicle-kilometres in the six wind intervals. In the daily summary, each counter–year–season accident total is distributed according to observed daily traffic allocated across three wind intervals by ten-minute wind frequency. Expected counts sum to observed counts within every stratum, and uncertainty is obtained from 5,000 bootstrap samples of whole road sections or counters. These are descriptive standardisations of accident-containing strata; the conditional Poisson models remain the inferential traffic analyses.

4 Results

The evidence is presented in three levels. First, weather-frequency O/E results use the broad 2007–2025 sample. Second, matched-time results check the weather comparison. Third, annual and daily traffic analyses use narrower linked samples to examine whether the mean-wind pattern remains after traffic is included. Raw counts and detailed checks are placed in the appendix.

4.1 Study Sample and Coverage

Table 4.1: Coverage of the retained analyses. One counter-day is one physical counter site on one date. A counter-day with daytime wind has a matched station within 20 km and at least one valid ten-minute mean-wind observation between 10:00 and 21:59; its traffic value is still the observed 24-hour total.

Analysis step	Retained	Starting set	Share
Wind match within 10 km	5,098	6,414	79.48%
Wind match within 20 km (primary)	6,259	6,414	97.58%
Wind match within 30 km	6,402	6,414	99.81%
Temperature match within 20 km	6,259	6,414	97.58%
Exact annual road-section match, 2007–2025	5,731	6,414	89.35%
Stratified accident-rate analysis	4,933	6,259	78.81%
Daily counter-days with daytime wind	763,171	774,274	98.57%
Allocated daily-counter rate, 2019–2024	762	1,863	40.90%
Full-day-mean daily-counter check, 2019–2024	767	1,863	41.17%

The 20 km threshold includes most accidents while limiting the spatial distance to the weather proxy. Increasing the threshold to 30 km adds 207 accidents but can be misleading in complex terrain. The road-section analysis is much narrower than the main analysis and is not equally representative.

The two annual-traffic rows have different analytical meanings. The first records an exact road-section and year link. The second also requires a defined traffic period and usable annual traffic and weather-frequency information. It is the population used for the estimated crash-rate analysis below.

The selection is explicit: 126,607 source accidents are reduced to 6,414 rural injury accidents by the outcome and location rules; 6,259 of these have qualifying wind and temperature matches. The two variables are matched independently, but the same station and ten-minute observation is selected for every matched accident in the current combined weather data. Of

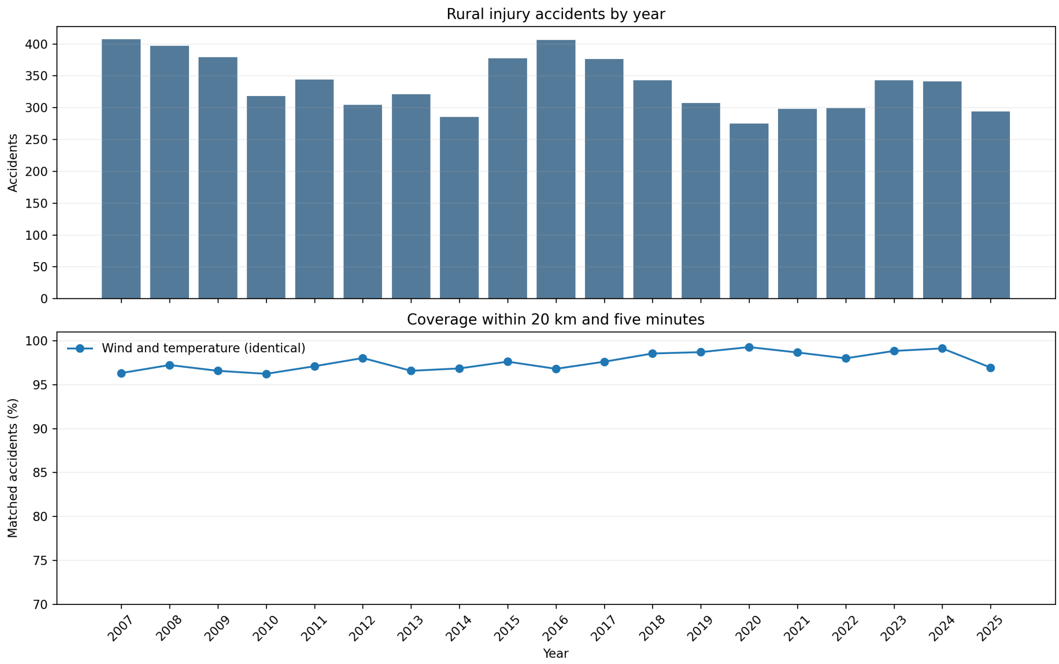

Figure 4.1: Annual rural injury-accident counts and the percentage matched to wind or temperature within 20 km and five minutes. The wind and temperature match counts are identical in every year; no missing temperature value is imputed.

these, 5,731 have an exact road-section/year traffic link and 4,933 enter the conditional traffic model. Within 2019–2024, 762 of 1,863 accidents enter the allocated daily-rate model after requiring the exact road section, a valid observed counter-day, the same weather station for the accident and denominator, and all 20 km distance limits. The less restrictive full-day-mean check includes 767 accidents and is reported only in the appendix. Each reduction answers a different data requirement, and no unmatched accident is silently reassigned.

4.2 Accident Frequency by Weather

Each figure in this section contains the complete year and the four seasons. The solid bars show all injury accidents; the hatched bars show the nested serious-or-fatal subset. Counts are printed above the bars. Seasonal estimates in the upper weather intervals are descriptive because several contain few accidents.

4.2.1 Mean Wind

Ratios are close to one across most low and moderate intervals. The plotted ratios increase from 15 m/s. At mean wind speeds of 20–25 m/s, 53 accidents were observed compared with 26.4 expected. The O/E ratio is 2.01, with a station-clustered interval of 1.44–2.63. At mean wind speeds of at least 25 m/s, 15 accidents were observed compared with 5.2 expected (O/E 2.88; 95% interval 0.97–6.02). This interval includes one and is too imprecise to support a separate conclusion at 25 m/s or above. The more informative upper mean-wind result is the 20–25 m/s interval. For the serious-or-fatal subset, the corresponding all-year O/E ratios are

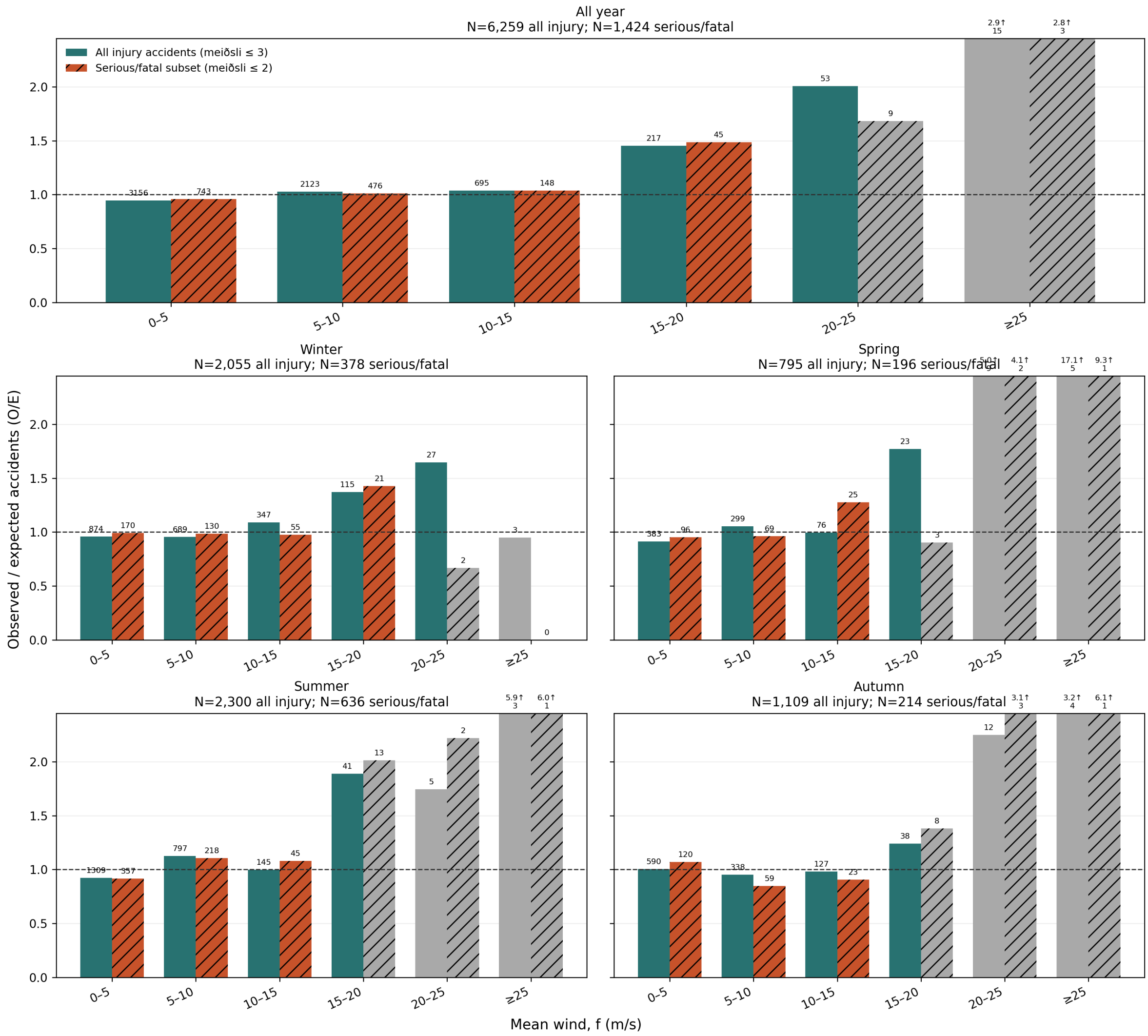

Figure 4.2: Mean-wind O/E for the complete year and four seasons, 2007–2025. All injury accidents (meiðsli ≤ 3) and their serious-or-fatal subset (meiðsli ≤ 2) are compared with local weather frequency. Grey bars contain fewer than 20 accidents; arrows mark ratios above the shared vertical scale. Confidence intervals are retained in the result table rather than drawn as whiskers.

1.68 at 20–25 m/s (9 accidents) and 2.82 at 25 m/s or above (3 accidents). The counts are too small for a precise comparison with all injury accidents.

The 5 m/s grouping is used throughout because it gives a direct, readable description of the upper wind intervals without implying that the boundaries are physical thresholds.

The upper-wind pattern is stable under the pre-defined spatial alternatives (Table 4.2). At 20–25 m/s, O/E is 2.21 using 10 km, 2.01 using the chosen 20 km limit, and 1.97 using 30 km. The wider radii add accidents but do not create the association. The ≥ 25 m/s interval remains imprecise under every radius because it contains only 15–17 accidents.

Table 4.2: Primary mean-wind O/E under three weather-station distance limits

Radius	Primary	Mean wind	Observed	Expected	O/E (95% interval)
10 km	No	15–20	190	123.9	1.53 (1.31–1.74)
10 km	No	20–25	49	22.2	2.21 (1.54–2.98)
10 km	No	≥25	15	4.6	3.28 (1.05–7.13)
20 km	Yes	15–20	217	149.2	1.45 (1.25–1.64)
20 km	Yes	20–25	53	26.4	2.01 (1.44–2.63)
20 km	Yes	≥25	15	5.2	2.88 (0.97–6.02)
30 km	No	15–20	218	152.3	1.43 (1.23–1.62)
30 km	No	20–25	53	26.9	1.97 (1.41–2.57)
30 km	No	≥25	15	5.4	2.80 (0.93–5.82)

4.2.2 Wind Gust

Matched-time wind gust is an additional weather measure rather than a daily maximum. Its O/E pattern rises through the upper intervals. At gusts of at least 35 m/s, 25 accidents were observed compared with 5.0 expected (O/E 5.01; 95% interval 2.95–7.48). This supports the interpretation that hazardous wind conditions are over-represented, but gust and mean wind are related measures and should not be treated as independent results. The serious-or-fatal subset has O/E 2.60 at 25–30 m/s (25 accidents) and 4.95 at 35 m/s or above (5 accidents); the latter estimate is very sparse.

Appendix Table A.1 gives the highest-gust result under three weather-station distance limits.

4.2.3 Temperature

Temperature is interpreted cautiously. Valid temperature was available for 6,259 accidents (97.58%). Coverage by year is shown in Figure 4.1, and no temperature value is imputed. After adjustment by temperature station and season, the estimates were non-monotonic. O/E was below one at 3–6°C (808 observed and 1,137.0 expected; O/E 0.71, station-bootstrap 95% CI 0.67–0.76), but above one at -3–0°C (O/E 1.22, 1.15–1.29). It also increased in the warm tail: at 12–15°C O/E was 1.43 (1.31–1.54), and at 15°C or above 248 accidents were observed compared with 114.2 expected (O/E 2.17, 1.90–2.45). This pattern may reflect travel behaviour or weather and road conditions associated with temperature. It is not interpreted as a causal temperature effect. For the serious-or-fatal subset, all-year O/E is 0.69 at 3–6°C and 2.24 at 15°C or above (176 and 69 accidents, respectively).

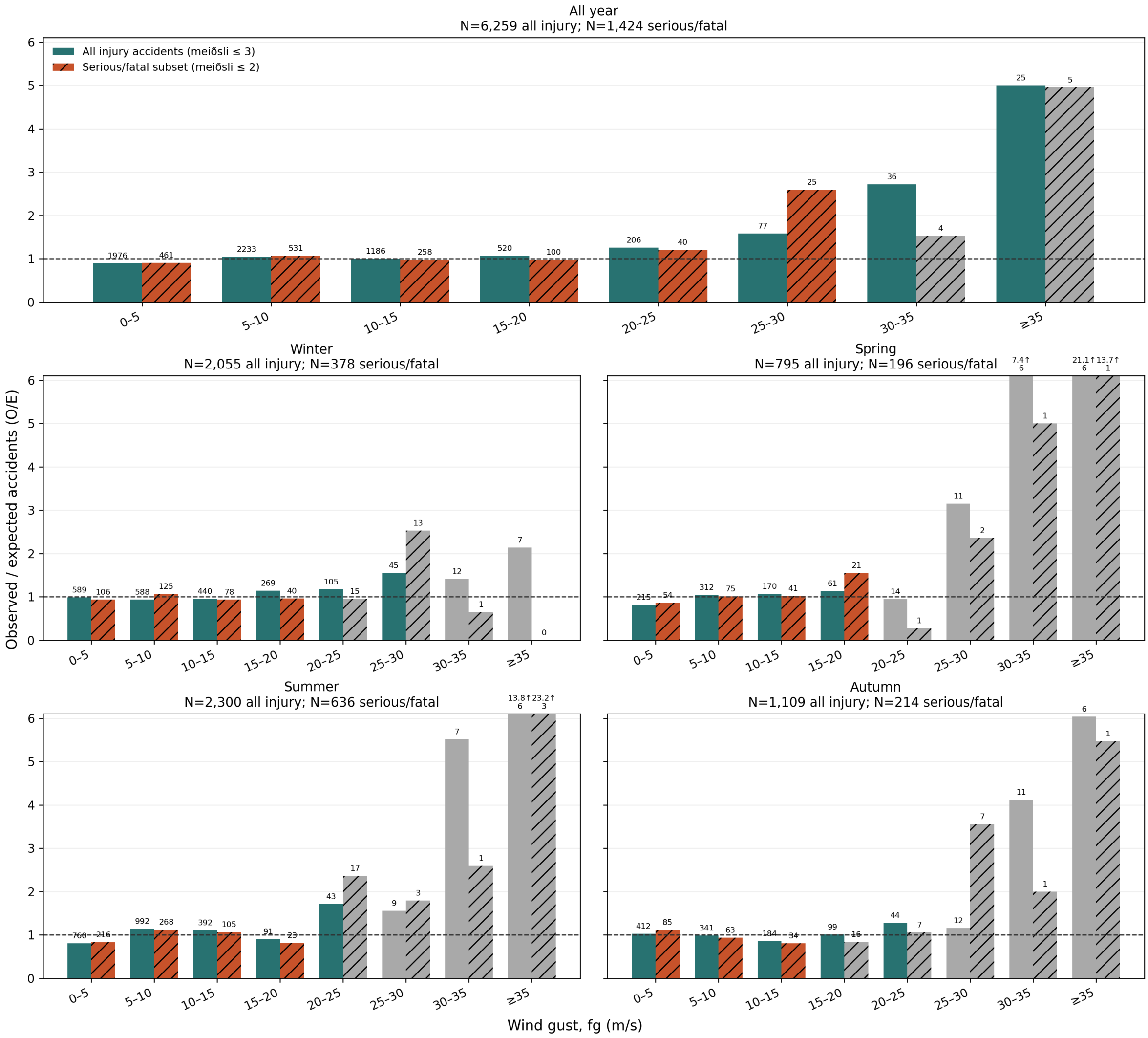

Figure 4.3: Wind-gust O/E for the complete year and four seasons. Gust is taken from the ten-minute observation matched to the accident time; it is not a daily maximum. The two outcome groups are nested. Grey bars contain fewer than 20 accidents, arrows mark ratios above the shared scale, and uncertainty intervals are retained in the result table.

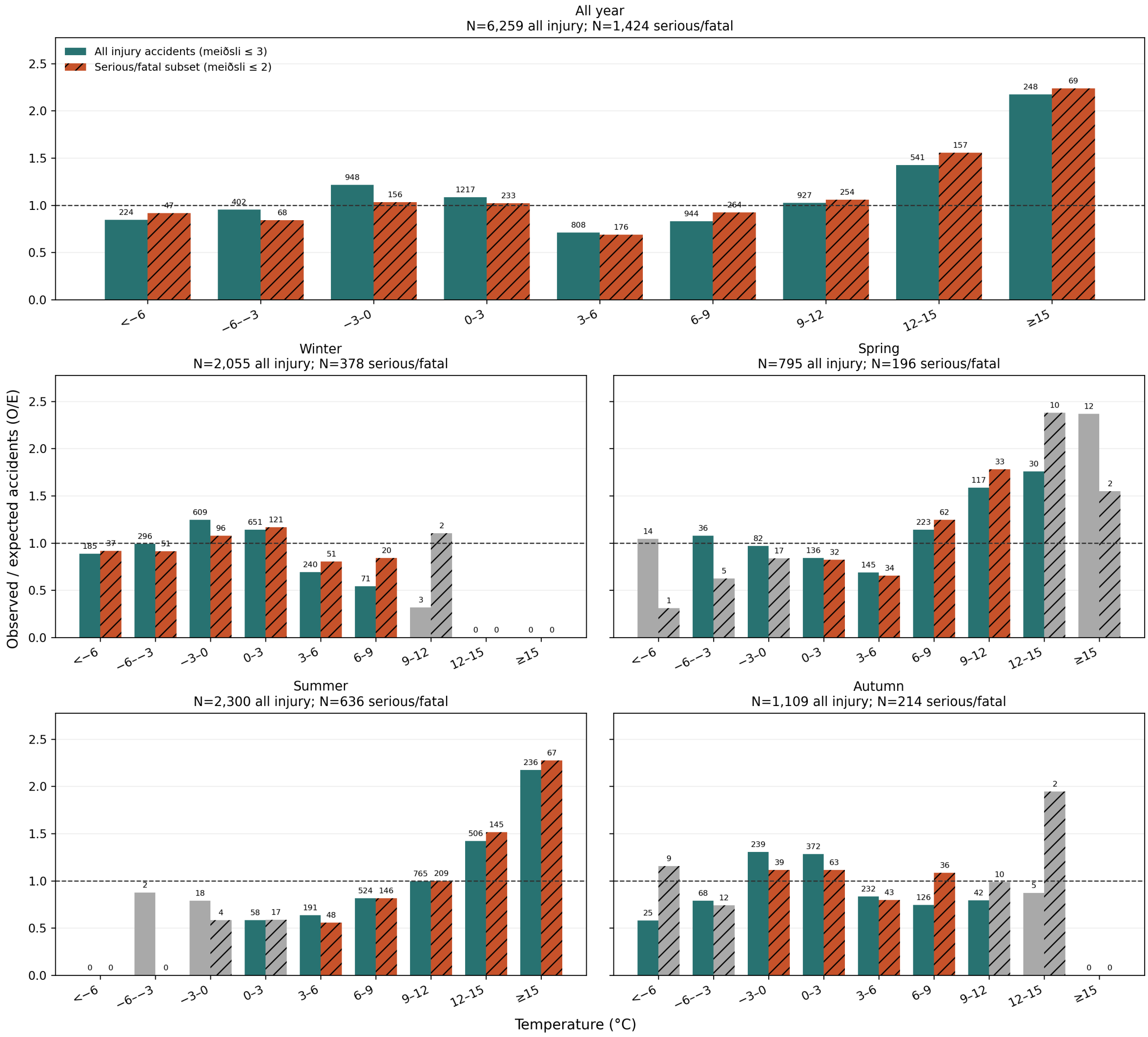

Figure 4.4: Temperature O/E for the complete year and four seasons, standardised by matched temperature station and study season. The two outcome groups are nested. Grey bars contain fewer than 20 accidents, arrows mark ratios above the shared scale, and uncertainty intervals are retained in the result table. The non-monotonic pattern is interpreted cautiously.

Because weather conditions and reporting coverage can change over a 19-year period, Table 4.3 repeats selected mean-wind and temperature results after defining the expected distribution within weather station, season, and year. The simpler pre-specified station–season result remains the main result.

Table 4.3: Selected pooled and year-adjusted O/E estimates

Variable	Interval	Observed	Station + season O/E (95% interval)	Station + season + year O/E (95% interval)
Mean wind (m/s)	15–20	217	1.45 (1.25–1.64)	1.45 (1.26–1.65)
Mean wind (m/s)	20–25	53	2.01 (1.44–2.63)	2.02 (1.46–2.62)
Mean wind (m/s)	≥25	15	2.88 (0.97–6.02)	2.94 (0.96–6.22)
Temperature (deg C)	–3–0	948	1.22 (1.15–1.29)	1.21 (1.14–1.28)
Temperature (deg C)	0–3	1,217	1.08 (1.02–1.14)	1.09 (1.03–1.15)
Temperature (deg C)	3–6	808	0.71 (0.67–0.76)	0.71 (0.67–0.76)

The results change little. At 20–25 m/s, O/E changes from 2.01 to 2.02. For temperature, the 3–6°C estimate remains 0.71, while the estimate at 15°C or above changes from 2.17 to 2.15.

4.3 Matched-Time Results

The case-crossover analysis includes 6,257 mean-wind and gust strata with 21,139 control times for each measure, and 6,257 temperature strata with 21,144 control times. A 5 m/s increase in mean wind is associated with a matched odds ratio of 1.09 (95% CI 1.05–1.13). The categorical estimate changes mainly at mean wind of at least 15 m/s: the odds ratio relative to 0–5 m/s is 1.61 (95% CI 1.38–1.87). This supports the high-wind pattern after exact control for station, year, month, weekday, and clock time.

For gust, the matched odds ratio per 5 m/s is 1.06 (95% CI 1.04–1.09). Relative to 0–10 m/s, it is 1.36 (1.15–1.60) at 20–25 m/s, 1.72 (1.30–2.28) at 25–30 m/s, and 2.74 (1.96–3.85) at 30 m/s or above. The matched-time result agrees with the upper-gust O/E pattern.

The temperature estimates are not monotonic. Relative to 0–3°C, the matched odds ratio is 1.13 (95% CI 1.02–1.25) at -3–0°C, 0.59 (0.53–0.65) at 3–6°C, and 0.99 (0.80–1.23) at 15°C or above. The matched-time result reproduces the dip at moderate temperatures but not the warm-tail increase in the O/E analysis. The methods therefore give different results at warm temperatures, which calls for caution.

When mean wind and temperature are included together, the matched odds ratio for mean wind of at least 15 versus 0–5 m/s is 1.68 (95% CI 1.44–1.96). This is close to the corresponding single-variable estimate of 1.61. The non-monotonic temperature pattern also remains. The small change in the wind estimate shows that including these temperature categories changes the upper-wind association very little.

4.4 Daylight, Season, and Severity

The raw distributions by hour, season, daylight, and temperature are shown in Appendix Figure A.7. They document the study sample and sparse categories but are not used to rank accident risk.

The matched-date daylight comparison holds location, month, weekday, and clock time fixed. Relative to daylight, the odds ratio is 1.14 (95% CI 0.88–1.49) for darkness and 1.10 (0.92–1.31) for civil twilight. Only 985 of the 6,414 strata change daylight class within their matched dates. The result provides no clear evidence of a daylight association and is much less informative than the wind analysis; it does not use vehicle counts.

Among the 6,259 accidents in the adjusted severity model, 1,424 are serious or fatal. Mean wind of at least 15 m/s has an odds ratio of 0.93 (95% CI 0.69–1.26) relative to 0–10 m/s. Darkness has an odds ratio of 0.89 (0.71–1.12) relative to daylight. Thus the data show a clear strong-wind pattern for accident occurrence, but not for severity conditional on a recorded injury accident. The complete adjusted estimates are shown in the appendix.

The formal matched-time wind-by-season comparison uses the same 6,257 mean-wind strata as the case-crossover analysis, with coarse intervals for stability. The omnibus interaction test gives $\chi^2(6) = 14.30$, $p = 0.026$. The estimated odds ratios for at least 15 versus 0–10 m/s are 1.44 in winter, 2.02 in spring, 2.53 in summer, and 1.37 in autumn. The omnibus test provides evidence that at least some seasonal wind associations differ, but it does not identify a particular pair of seasons as different. Season-specific estimates are therefore not ranked.

4.5 Traffic Results

This section first gives the broader annual-traffic comparison and then the higher-resolution daily-counter analysis. Annual data cover more accidents and include road length, but traffic within a season is allocated by weather frequency. Daily data start from observed 24-hour counts, but cover fewer locations and still require an estimated split among within-day wind intervals. Neither source directly observes traffic at the accident minute.

4.5.1 Broader Annual-Traffic Comparison

The descriptive estimated rate increased from 8.45 rural injury accidents per 100 million estimated vehicle-kilometres at 0–5 m/s to 40.46 at 20–25 m/s. After comparison within road section, year, and traffic period, the corresponding conditional rate ratio was 2.27. The descriptive rate and conditional estimate differ because they aggregate and control road-section and period differences in different ways; both use traffic allocated to wind intervals rather than traffic observed at ten-minute resolution.

The conditional Poisson model includes 4,933 accidents in 3,984 road-section–year–traffic-period groups that contain accidents. Relative to 0–5 m/s, the estimated within-stratum rate ratio is 1.63 (95% CI 1.39–1.90) at 15–20 m/s, 2.27 (1.68–3.07) at 20–25 m/s, and 4.95 (3.07–7.98) at 25 m/s or above. The uppermost estimate contains 18 accidents and is less precise. These rates use traffic allocated within each period according to local wind frequency;

they are not directly observed 10-minute traffic.

Applying the same estimated vehicle-kilometre construction to temperature includes 4,921 accidents in 3,976 road-section–year–traffic-period groups. Relative to 0–3°C, the within-group rate ratio is 0.79 (95% CI 0.68–0.93) below -6°C and 0.60 (0.55–0.67) at 3–6°C. It increases to 1.34 (1.15–1.55) at 12–15°C and 2.02 (1.68–2.44) at 15°C or above. The warm-end pattern agrees with the O/E result, but not with the matched-time result. Only stations used by the road traffic model are allowed, and vehicle-kilometres are allocated according to temperature frequency rather than observed directly at each temperature.

The serious-or-fatal model includes 1,055 accidents. Relative to 0–5 m/s, its rate ratio is 1.92 (95% CI 1.40–2.64) at 15–20 m/s and 2.23 (1.14–4.37) at 20–25 m/s. The at-least-25 m/s estimate is 4.91 (1.76–13.64), but contains only four accidents and is correspondingly imprecise.

Using coarse seasonal intervals, the rate ratio for at least 15 versus 0–10 m/s is 1.46 (95% CI 1.20–1.77) in winter, 2.97 (2.06–4.28) in spring, 2.20 (1.61–3.01) in summer, and 1.39 (1.02–1.89) in autumn. The corresponding upper-category counts are 123, 34, 44, and 46. The pattern is not confined to winter, although spring and autumn use derived VHDU traffic and should not be compared as precise seasonal rankings.

For serious or fatal accidents, the corresponding at-least-15 m/s estimates are 1.73 in winter, 1.95 in spring, 3.00 in summer, and 1.90 in autumn, based on 25, 6, 15, and 12 accidents. The spring upper category contains only six accidents and is imprecise. The detailed figure is shown in the appendix rather than used to rank seasons.

In separate coarse models, the rate ratio for at least 15 versus 0–10 m/s is 1.42 (95% CI 1.19–1.68) for one-vehicle accidents and 2.37 (1.90–2.97) for accidents involving two or more vehicles. The corresponding upper-category counts are 150 and 97. These separately fitted subgroup estimates do not by themselves prove that the associations differ, and the larger multiple-vehicle estimate should not be given a mechanical explanation that the available accident records cannot test. The complete comparison is shown in the appendix.

4.5.2 Daily-Counter Traffic Comparison

Observed traffic decreases as strong wind persists. Relative to the calendar expectation at the same counter, traffic is 96.1% on days with more than zero but less than two hours at $f \geq 15$ m/s, 93.8% with 2–6 hours, and 86.8% with at least six hours. This duration measure is more informative than a single daily mean, but still describes selected counter sites.

The allocated daily-counter model includes 762 accidents in 491 counter–year groups that contain accidents. Relative to 0–10 m/s, the rate ratio is 1.61 (95% CI 1.30–2.00) at 10–15 m/s and 3.62 (2.65–4.94) for at least 15 m/s. The latter category contains 47 accidents. The calculation starts from observed daily traffic, but the within-day split is estimated from the counter station's ten-minute wind frequency; it is not observed hourly traffic. Accident-time mean wind is taken from that same station, which is required to be within 20 km of the accident as well as the assigned counter. For serious or fatal accidents, the corresponding estimates are 1.65 (1.05–2.60; 23 accidents) at 10–15 m/s and 4.53 (2.50–8.20; 13 accidents) for at least 15 m/s. Restricting the allocation and accident times to 07:00–24:00 changes the all-injury upper estimate only modestly, from 3.62 (2.65–4.94; 47 accidents) to 3.74

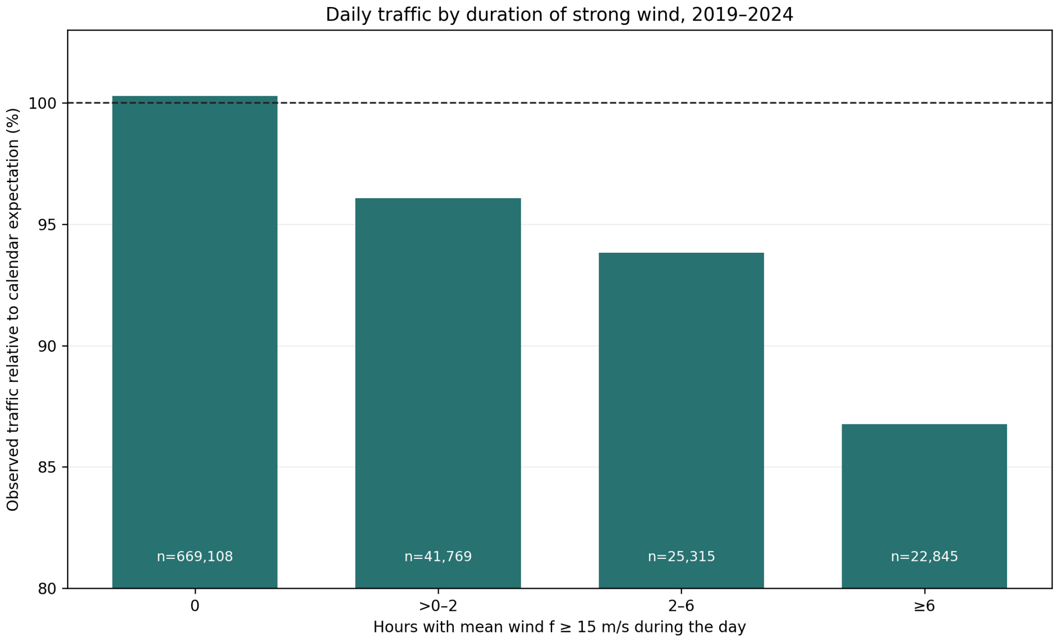

Figure 4.5: Observed daily traffic by duration of strong wind. Values are relative to the mean for the same counter, year, month, and weekday; labels show counter-days. Confidence intervals are reported in the result table.

(2.72–5.15; 44 accidents). This stability does not validate the assumed time distribution; it only shows that the headline direction is not created by including midnight–07:00 weather.

The same allocated daily data give the seasonal comparison in Figure 4.7. Its at-least-15 m/s category contains 24 winter, 10 spring, 7 summer, and 6 autumn accidents, so it is not used to rank seasons. The overall six-degree-of-freedom interaction test gives $\chi^2(6) = 12.59$, $p = 0.050$. The narrower secondary test of only the at-least-15 m/s seasonal terms gives $\chi^2(3) = 8.77$, $p = 0.032$. Together with the larger annual-traffic result, this supports a possible seasonal difference but does not identify which pair of seasons differs.

Appendix Table A.11 addresses the two most important data questions. Restricting the rate model to the official VDU and SDU periods reduces the upper estimates but leaves their direction unchanged. Excluding daily counter records reported as zero changes the 20–25 m/s traffic estimate from 68.9% to 69.3%, so those records do not explain the observed decline.

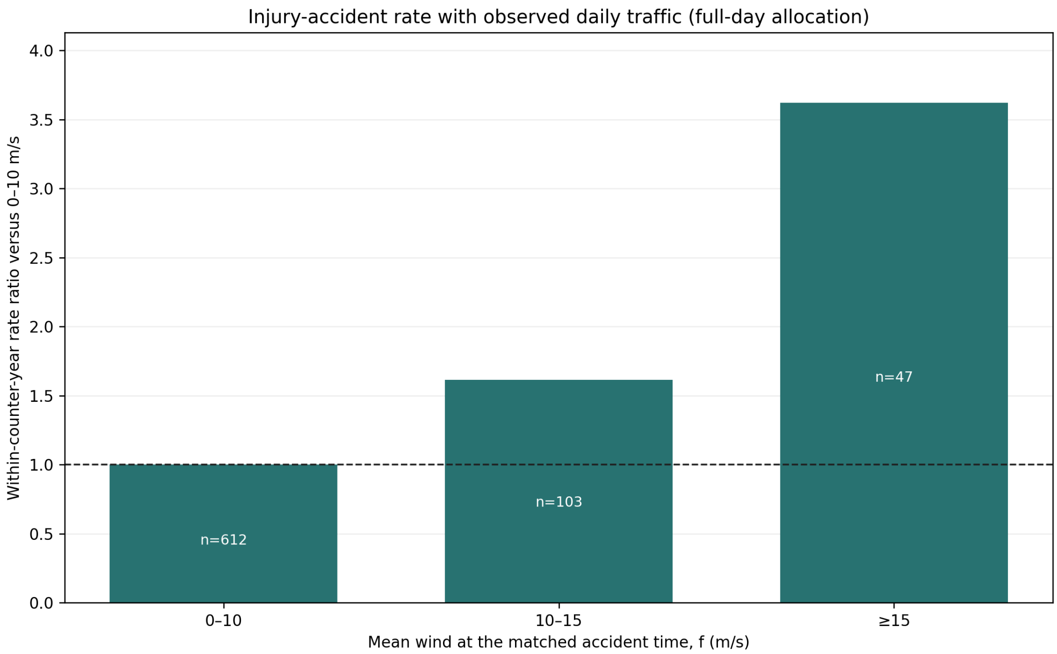

Figure 4.6: Daily-counter rural injury-accident rate ratios by accident-time mean wind. Observed 24-hour traffic is allocated within each day in proportion to the counter station's ten-minute wind frequency, and accidents use that same station. Comparisons are within counter and year; labels show accidents. Confidence intervals are reported in the text and result tables.

Table 4.4: Summary of the evidence used to interpret the main result

Analysis	Comparison	Estimate (95% interval)	Interpretation and principal limitation
Primary O/E	20–25 m/s	2.01 (1.44–2.63)	Accident occurrence relative to local wind frequency; traffic is not included.
Matched time	$\geq$ 15 vs 0–5 m/s	OR 1.61 (1.38–1.87)	Same calendar time; no direct measure of unusual daily travel changes.
Road-section traffic	20–25 vs 0–5 m/s	RR 2.27 (1.68–3.07)	Within road, year and traffic period; traffic is estimated from annual values.
Allocated daily rate	$\geq$ 15 vs 0–10 m/s	RR 3.62 (2.65–4.94)	Observed daily total with estimated within-day allocation; 762 linked accidents.

The four rows estimate different quantities and should not be compared by magnitude. Together they show the same direction: accidents are over-represented in strong wind even though observed traffic tends to be lower on windy days. The appendix direction check applies the daily percentages mechanically and changes the annual-model 20–25 m/s rate ratio from 2.27 to 3.32. This is not an adjusted estimate because a full-day count cannot identify traffic during individual ten-minute observations; it only shows that the proportional allocation is unlikely to create the elevated rate ratio.

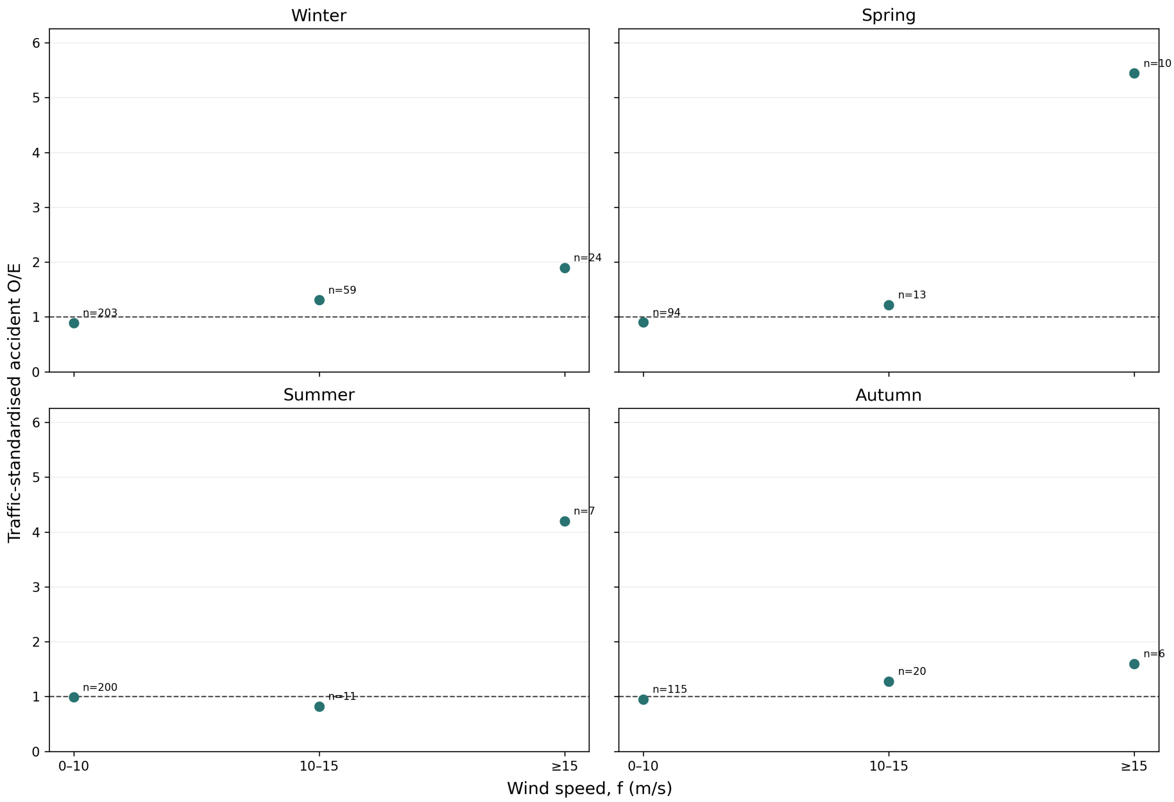

Figure 4.7: Traffic-standardised rural injury-accident occurrence by mean wind and season at selected daily counters, 2019–2024. Daily totals are observed, but the vehicles in each wind interval are estimated by allocating every daily total according to the fraction of valid ten-minute observations in that interval. Labels show observed accidents. All panels use the same vertical scale; uncertainty intervals are retained in the result table.

The appendix gives the raw hour, season, daylight, and temperature counts and the detailed subgroup and traffic checks. These figures describe the sample or show how stable the main finding is; they are not additional estimates of hourly accident risk. The available traffic files do not contain the hourly counts needed for a traffic-adjusted hour or daylight curve.

5 Discussion

5.1 Interpretation

Strong mean wind is over-represented at the times of rural injury accidents relative to its local frequency. Results are standardised within weather station and season, and uncertainty is clustered by station. The result describes association rather than a per-vehicle causal effect. Moderate wind intervals fluctuate around one, while several upper intervals contain few accidents.

Traffic remains important for interpretation because the main curve cannot separate risk per vehicle from the number of vehicles exposed. The annual and daily analyses point in the same direction: observed traffic is lower when strong wind persists, while both traffic-based accident models give rate ratios above one. Both have important limitations. The annual model allocates seasonal traffic by wind frequency, and the selected-counter model starts from measured 24-hour traffic but estimates its distribution within the day. The latter covers 762 accidents. These results agree with the main finding but are not a national estimate based on traffic observed at each accident time.

The additional weather results are informative but less simple than the main wind result. Gust shows a clear upper-tail pattern in both O/E and matched-time comparisons, although it is closely related to mean wind. Temperature has good overall independent-match coverage but a non-monotonic result across intervals and methods. Both the O/E and estimated vehicle-kilometre analyses show an increase above 12°C, whereas the matched-time comparison does not. The wind estimate changes little when temperature is included in the same matched model. The matched daylight estimates are close to one with wide intervals. The adjusted time, daylight, and season comparisons concern injury severity, not accident occurrence. The severity model does not show higher serious-or-fatal severity at strong wind, which distinguishes occurrence from consequences among accidents. The formal seasonal interaction test is suggestive but does not provide strong evidence that the wind association differs by season.

The direction of the main result agrees with earlier studies comparing accident shares with the time spent in adverse weather Edwards (1994) and with the engineering evidence that cross-wind can impair vehicle stability Baker (1986, 1994). Its contribution is a longer Icelandic injury- accident series, local station-and-season standardisation, and independent matched-time and traffic checks rather than a new physical wind threshold. The non-monotonic temperature estimates are consistent with previous evidence that temperature associations can vary by range and season Lee et al. (2014); Zhan et al. (2020). The fall in daily traffic during sustained wind is also consistent with recorder-based evidence that adverse weather changes travel demand Sathiaraj et al. (2018). Earlier work helps interpret the results but does not remove the study's measurement and traffic limitations.

The serious-or-fatal all-year curves broadly follow the all-injury curves. Seasonal panels are less stable and are used to show where the sample becomes sparse, not to rank seasons. In the weather-frequency mean-wind result, only three serious-or-fatal accidents occur at 25 m/s or above. This precludes a precise comparison of effect sizes across outcomes or seasons. The coarse traffic models also show upper-wind rate ratios above one for both vehicle-count groups. The estimate is larger for multiple-vehicle accidents, but separately fitted subgroup estimates are not a formal comparison and do not identify the underlying collision mechanism.

The O/E and traffic-model estimates are not competing versions of the same number. O/E compares accident occurrence with the amount of time each wind interval is observed, using 6,259 matched accidents. The conditional model compares estimated accident rates within road section, year, and traffic period and uses 4,933 accidents with the required traffic links and positive estimated vehicle-kilometres. Its larger upper-bin estimate is compatible with traffic falling during strong wind, but it also depends on allocated rather than directly observed 10-minute traffic. The agreement in direction is more informative than the difference in magnitude.

5.2 Strengths

The main strengths are the large 2007–2025 accident sample, 10-minute weather resolution, explicit spatial coverage reporting, local wind-frequency standardisation, and station-clustered bootstrap intervals. A separately specified time-stratified analysis controls station and recurring calendar time, while the traffic analyses examine the limitation caused by unavailable traffic at the accident time. The matching data include unmatched accidents and record the reason for each unsuccessful match. The fixed hierarchy keeps the connection between the research question, method, and result clear.

The empirical contribution is a quantified comparison of Icelandic rural injury accidents with local weather frequency and the available traffic data. The computational contribution is an auditable pipeline that preserves source identifiers, records exclusions and match coverage, separates large source preparation from compact analysis CSV files, and generates the reported tables and figures from code. Automated validation checks the main sample sizes, definitions, and numerical results. This implementation makes the analytical claims inspectable without treating software organisation itself as a research result.

5.3 Limitations

- The main result is not traffic-adjusted. It should not be interpreted as accident risk per vehicle-kilometre.
- The case-crossover analysis controls recurring calendar time but not unusual traffic, road conditions, visibility, or closures on a particular day. Its ≥ 15 m/s category summarises the upper-wind pattern rather than a physical threshold.
- A weather station up to 20 km away is a proxy for conditions at the accident location. Terrain can produce important differences over shorter distances.

- The stratified crash-rate analysis represents 4,933 accidents and uses annual seasonal traffic allocated by wind frequency, rather than traffic observed at the accident minute. It aligns accident and road-section weather to the same station, but excludes accidents for which that station is farther than 20 km from the accident.
- The temperature vehicle-kilometre model uses 4,921 accidents with valid temperature at the road-assigned weather station. This is less than the independently matched temperature sample because the numerator and denominator are required to use the same traffic-model station.
- Daily counter data cover 2019–2024. The duration analysis relates the observed 24-hour total to hours of strong wind; it does not observe traffic separately during those hours.
- Daily traffic counts cover selected counter sites. Multiple counters can represent different directions, lanes, or locations; keeping counters separately avoids double counting but does not provide national coverage.
- The allocated daily-rate model includes 762 accidents with an exact road-section counter link and valid counter-day. It assumes the observed daily traffic is distributed across wind intervals in proportion to valid ten-minute observations. This is an estimated within-day split, not observed hourly traffic. The accident-time wind and denominator now use the same station, within 20 km of both the accident and counter, but the accident vehicle need not have passed the assigned counter.
- The 07:00–24:00 daily check assumes zero traffic outside that interval and uniform traffic within it. This is deliberately an assumption check, not a reconstruction of hourly traffic.
- Estimated rates are restricted to rural injury accidents on linked road sections. They should not be compared directly with broad national or international accident rates unless the outcome and vehicle-kilometre traffic quantity are defined in the same way.
- The highest wind intervals contain relatively few accidents. Several intervals and comparison results are shown, so individual estimates in the upper tail should be interpreted with caution.
- Fixed 2020–2024 urban boundaries are applied to the entire 2007–2025 accident period.
- The analysis is observational. Unmeasured road condition, vehicle type, speed, visibility, and driver behaviour can contribute to the pattern.

5.4 Further Work

The most valuable additional data would be verified hourly traffic counts with counter coordinates, stable road-section identifiers, lane and direction metadata, and documented downtime. Linking those counts to the same weather stations would provide genuine hourly traffic for accident and control times. Daily totals cannot recover that information, so further modelling of the present totals would not solve this limitation. With hourly data, a future analysis could distinguish vehicle classes, wind direction relative to the road, and road geometry. Better traffic data would add more value than additional models based on the current information.

6 Conclusion

Question 1 asked for weather-frequency-standardised estimates. Of 6,414 rural injury accidents during 2007–2025, 6,259 were matched to valid wind within 20 km and five minutes. At 20–25 m/s, 53 accidents were observed compared with 26.4 expected, giving O/E 2.01 with a 95% station-clustered interval of 1.44–2.63. The sparse category at 25 m/s or above had O/E 2.88, but its interval of 0.97–6.02 prevents a separate precise conclusion. The serious-or-fatal subset followed the same broad mean-wind pattern but was too sparse for firm seasonal or upper-tail comparisons: it contained nine accidents at 20–25 m/s and three at 25 m/s or above.

Accident-time gust showed a clear upper-tail pattern: at 35 m/s or above, O/E was 5.01, and the matched odds ratio at 30 m/s or above was 2.74 relative to 0–10 m/s. Temperature was available for 6,259 accidents and was non-monotonic. The matched mean-wind odds ratio for at least 15 versus 0–5 m/s changed from 1.61 to 1.68 when temperature was included. Temperature O/E was 2.17 at 15°C or above, and the estimated vehicle-kilometre model gave a rate ratio of 2.02 relative to 0–3°C. The matched-time estimate for the same warm interval was 0.99 with an interval including one. The warm-temperature result depends on the method and should be interpreted cautiously.

Question 2 asked whether matched-time comparisons support the weather-frequency findings. They support the upper-wind and upper-gust patterns: the mean-wind odds ratio was 1.61 for at least 15 versus 0–5 m/s, and the gust odds ratio was 2.74 for at least 30 versus 0–10 m/s. The methods did not agree at warm temperatures, so that result remains uncertain. Daylight and severity estimates are reported as context rather than answers to a separate main question.

Question 3 asked what the available traffic data add. The annual-traffic model gave a rate ratio of 2.27 at 20–25 m/s relative to 0–5 m/s. At selected daily counters, traffic fell to 86.8% of its calendar expectation on days with at least six hours of mean wind at or above 15 m/s. The allocated daily-counter model gave a rate ratio of 3.62 at 15 m/s or above relative to 0–10 m/s. These traffic estimates have the same direction as the main result, but traffic within each wind interval is estimated rather than observed directly. The seasonal daily-counter cells at at least 15 m/s contain 24, 10, 7, and 6 accidents in winter, spring, summer, and autumn. The full interaction test gave $p = 0.050$, while the narrower test of the upper-wind seasonal terms gave $p = 0.032$. Both are reported because the sparse data do not support selecting or ranking a single seasonal result.

The main conclusion is a quantified association between strong mean wind and higher relative occurrence of rural injury accidents in Iceland. It is not a causal estimate or a complete national risk per vehicle. Better linked hourly traffic data are required to estimate directly observed road risk per vehicle.

Bibliography

Ahmed, M. M., Abdel-Aty, M., and Yu, R. (2012). Assessment of interaction of crash occurrence, mountainous freeway geometry, real-time weather, and traffic data. *Transportation Research Record*, 2280(1), 51–59. https://doi.org/10.3141/2280-06.

Andrey, J., Mills, B., Leahy, M., and Suggett, J. (2003). Weather as a chronic hazard for road transportation in Canadian cities. *Natural Hazards*, 28, 319–343.

Baker, C. J. (1986). A simplified analysis of various types of wind-induced road vehicle accidents. *Journal of Wind Engineering and Industrial Aerodynamics*, 22(1), 69–85. https://doi.org/10.1016/0167-6105(86)90012-7.

Baker, C. J. (1994). The quantification of accident risk for road vehicles in cross winds. *Journal of Wind Engineering and Industrial Aerodynamics*, 52, 93–107. https://doi.org/10.1016/0167-6105(94)90041-8.

Edwards, J. B. (1994). Wind-related road accidents in England and Wales 1980–1990. *Journal of Wind Engineering and Industrial Aerodynamics*, 52, 293–303. https://doi.org/10.1016/0167-6105(94)90055-8.

Hagstofa Íslands. (2026). Urban-locality geographic data, 2020–2024. https://gis.is/geoserver/Hagstofan/ows.

Hrafnkelsdóttir, E. (2020). *Effects of wind on vehicles: Development of a calculation model* (in Icelandic). Vegagerðin research report 1800-707-2. https://www.vegagerdin.is/vegagerdin/gagnasafn/skyrslur/utgefnar-rannsoknaskyrslur/umferd/2020-umferd/ahrif-vinds-a-farartaeki-throun-reiknilikans/.

Johansson, Ö., Wanvik, P. O., and Elvik, R. (2009). A new method for assessing the risk of accident associated with darkness. *Accident Analysis & Prevention*, 41(4), 809–815. https://doi.org/10.1016/j.aap.2009.04.003.

Janes, H., Sheppard, L., and Lumley, T. (2005). Case-crossover analyses of air pollution exposure data: Referent selection strategies and their implications for bias. *Epidemiology*, 16(6), 717–726. https://doi.org/10.1097/01.ede.0000181315.18836.9d.

Lord, D. and Mannering, F. (2010). The statistical analysis of crash-frequency data: a review and assessment of methodological alternatives. *Transportation Research Part A*, 44(5), 291–305. https://doi.org/10.1016/j.tra.2010.02.001.

Lee, W.-K., Lee, H. A., Hwang, S.-S., Kim, H., Lim, Y.-H., Hong, Y.-C., Ha, E.-H., and Park, H. (2014). A time series study on the effects of cold temperature on road traffic injuries in Seoul, Korea. *Environmental Research*, 132, 290–296. https://doi.org/10.1016/j.envres.2014.04.019.

Theofilatos, A., and Yannis, G. (2014). A review of the effect of traffic and weather characteristics on road safety. *Accident Analysis & Prevention*, 72, 244–256. https://doi.org/10.1016/j.aap.2014.06.017.

Peng, R. D. (2011). Reproducible research in computational science. *Science*, 334(6060), 1226–1227. https://doi.org/10.1126/science.1213847.

Pu, Y., Chen, F., Chen, P., and Pan, X. (2018). Wind data collection and analysis of topographical features along a highway for traffic safety assessment based on mobile mapping technology. *Transportation Research Record*, 2672(42), 292–301. https://doi.org/10.1177/0361198118788465.

Samgöngustofa. (2026). Traffic accident statistics. https://www.samgongustofa.is/umferd/tolfraedi/umferdarslys/.

Sathiaraj, D., Punkasem, T.-O., Wang, F., and Seedah, D. P. K. (2018). Data-driven analysis on the effects of extreme weather elements on traffic volume in Atlanta, Georgia, USA. *Computers, Environment and Urban Systems*, 72, 212–220. https://doi.org/10.1016/j.compenvurbsys.2018.06.012.

Snæbjórnsson, J. Th., Baker, C. J., and Sigbjórnsson, R. (2007). Probabilistic assessment of road vehicle safety in windy environments. *Journal of Wind Engineering and Industrial Aerodynamics*, 95(9–11), 1445–1462. https://doi.org/10.1016/j.jweia.2007.02.020.

Veðurstofa Íslands. (2026). Weather observations and data services. https://api.vedur.is/weather/.

Vegagerðin. (2026). Traffic statistics. https://www.vegagerdin.is/samgongukerfid/vegakerfid/umferdartolur.

Washington, S., Karlaftis, M., Mannering, F., and Anastasopoulos, P. (2020). *Statistical and Econometric Methods for Transportation Data Analysis*. CRC Press.

Xing, F., Huang, H., Zhan, Z. Y., Zhai, X., Ou, C., Sze, N. N., and Hon, K. K. (2019). Hourly associations between weather factors and traffic crashes: Non-linear and lag effects. *Analytic Methods in Accident Research*, 24, 100109. https://doi.org/10.1016/j.amar.2019.100109.

Zhan, Z.-Y., Yu, Y.-M., Chen, T.-T., Xu, L.-J., and Ou, C.-Q. (2020). Effects of hourly precipitation and temperature on road traffic casualties in Shenzhen, China (2010–2016): A time-stratified case-crossover study. *Science of the Total Environment*, 720, 137482. https://doi.org/10.1016/j.scitotenv.2020.137482.

Young, R. K., and Liesman, J. (2007). Estimating the relationship between measured wind speed and overturning truck crashes using a binary logit model. *Accident Analysis & Prevention*, 39(3), 574–580. https://doi.org/10.1016/j.aap.2006.10.002.

A Supporting Analyses

These figures describe the study population, check the main wind pattern, or examine the traffic data. Unless stated otherwise, they are descriptive; sparse bars are shown in grey and their observed accident counts are printed on the bars.

A.1 Accident Characteristics

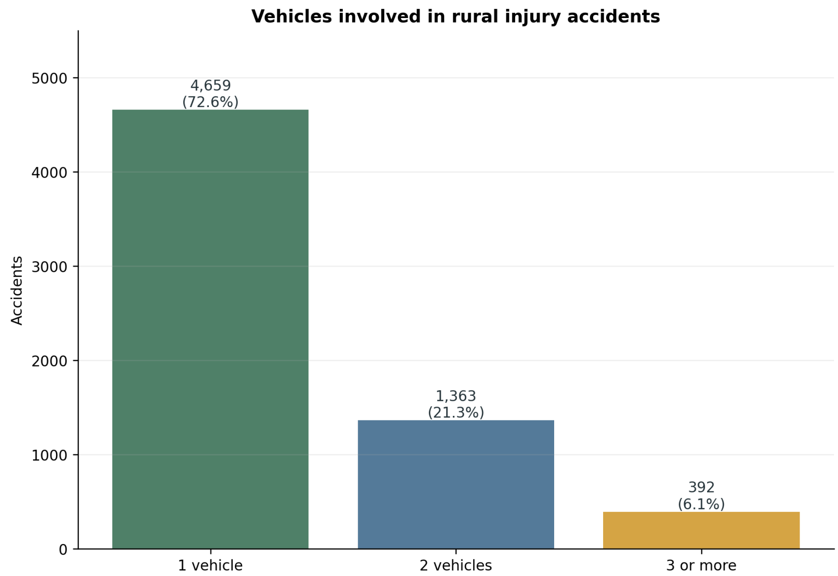

Figure A.1: Number of registered vehicles involved in the 6,414 rural injury accidents. Most selected accidents involved one registered vehicle.

A.2 Wind Gust Distance Check

Table A.1: Highest-gust result under three weather-station distance limits

Maximum distance	Matched accidents	Observed at ≥ 35	O/E	95% interval
10 km	5,098	25	5.83	3.36–8.79
20 km (primary)	6,259	25	5.01	2.95–7.48
30 km	6,402	25	4.90	2.81–7.33

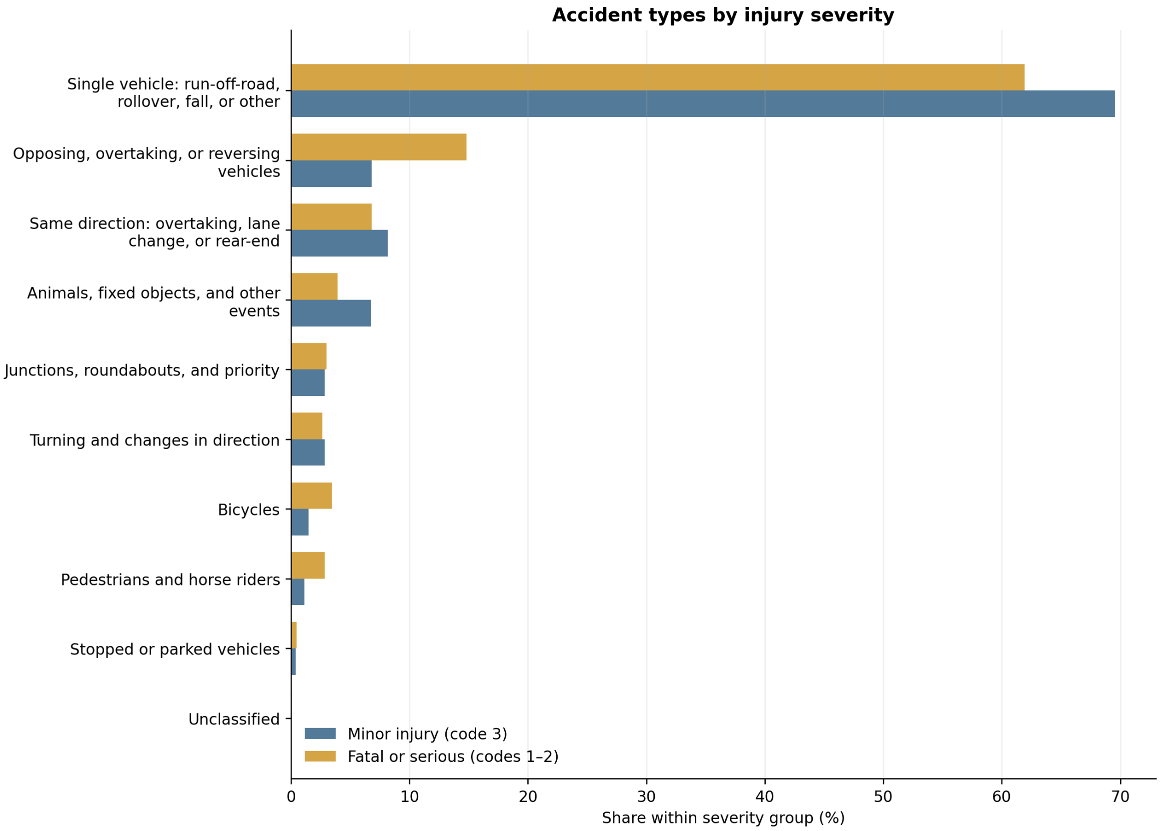

Figure A.2: Accident-type composition in minor-injury accidents and in the non-overlapping fatal-or-serious group. Percentages sum to 100 within each severity group.

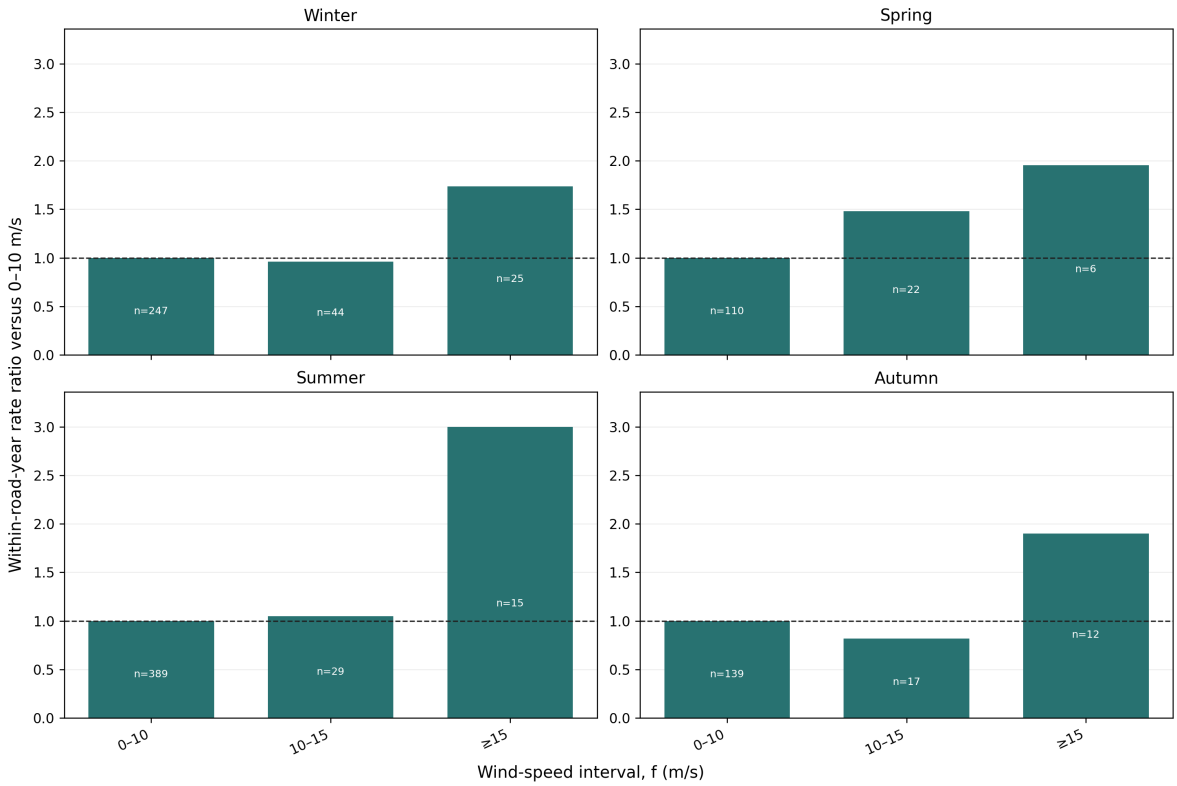

Figure A.3: Estimated serious-or-fatal accident rate ratios by season using coarse mean-wind intervals. The spring upper category contains six accidents, so the estimates show direction and uncertainty rather than precise differences between seasons.

A.3 Mean-Wind Subgroups

A.4 Raw Condition Counts, Temperature by Season, and Adjusted Severity

A.5 Daily-Counter Validation

Table A.2: Counter-location construction and independent coordinate check

Check	Result
Counter-site years in analysis location table	2,332
Counter-site years located from road geometry	2,325
Distinct located counter sites	473
Sites matched to an official 20 m road-station point within 10 m	435
Median / 90th-percentile coordinate difference	5.0 / 9.0 m

Coordinates are interpolated from the reported road station along official road geometry and remain labelled as estimates. The separate 20 m point match tests the interpolation; it does not replace the estimated location.

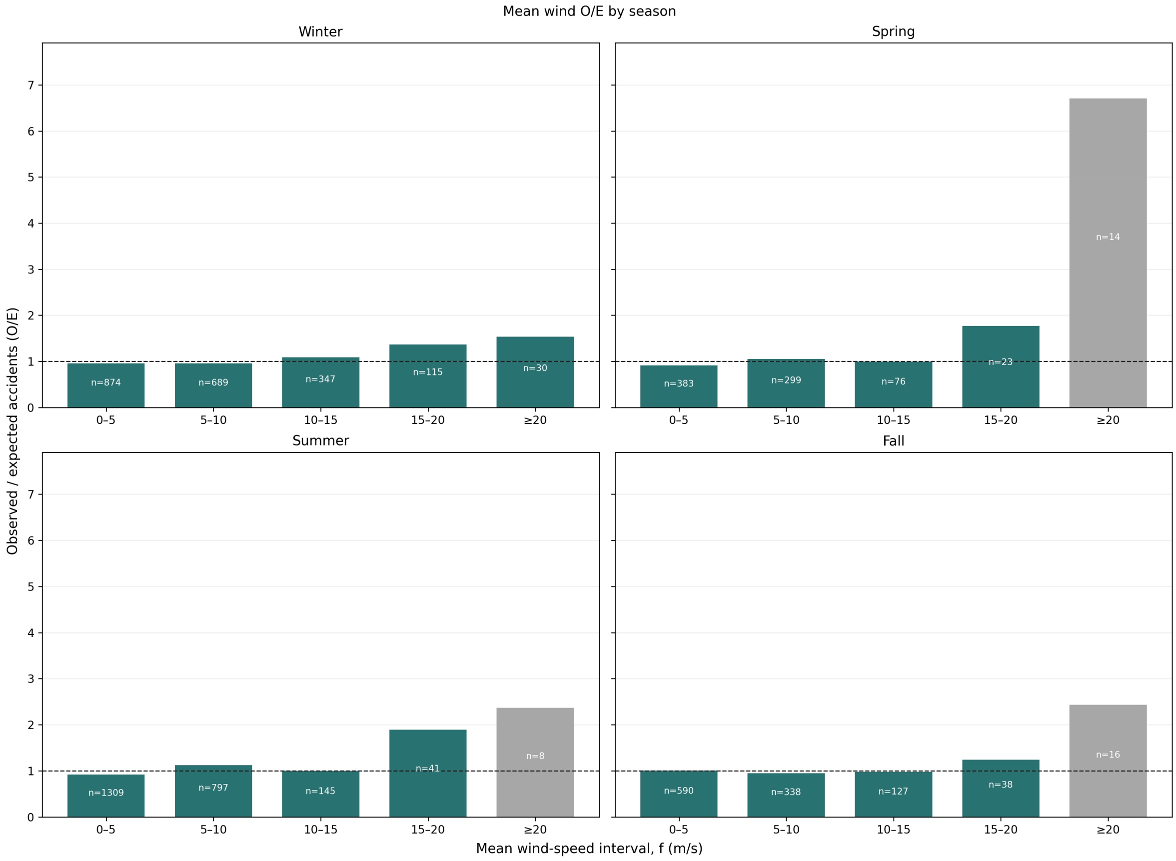

Figure A.4: Mean-wind O/E estimates by study season. These estimates are shown as a descriptive subgroup analysis; high-wind bins are sparse after the sample is divided by season. All panels share one vertical scale; grey bars contain fewer than 20 accidents.

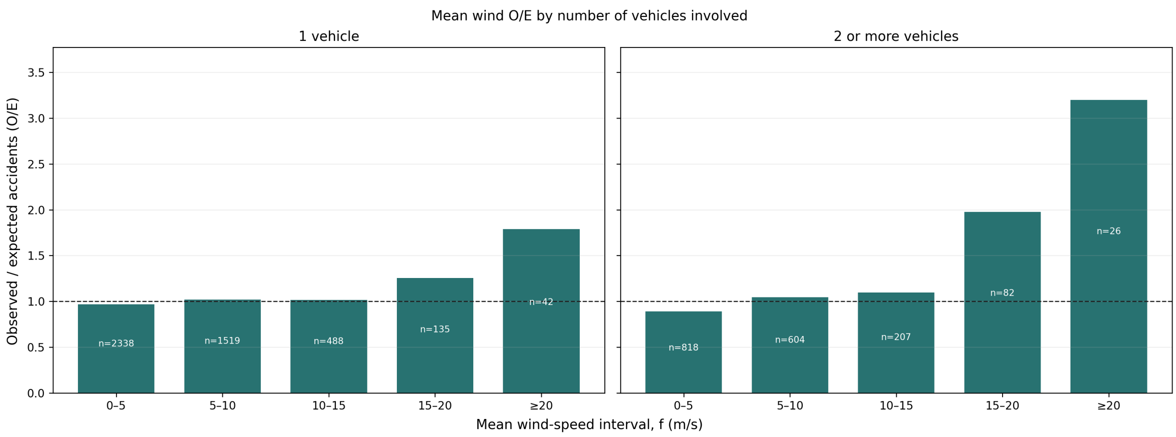

Figure A.5: Mean-wind O/E estimates by the number of registered vehicles involved. This is a descriptive subgroup analysis; it does not establish accident type or vehicle count as an explanatory variable. Both panels share one vertical scale; grey bars contain fewer than 20 accidents.

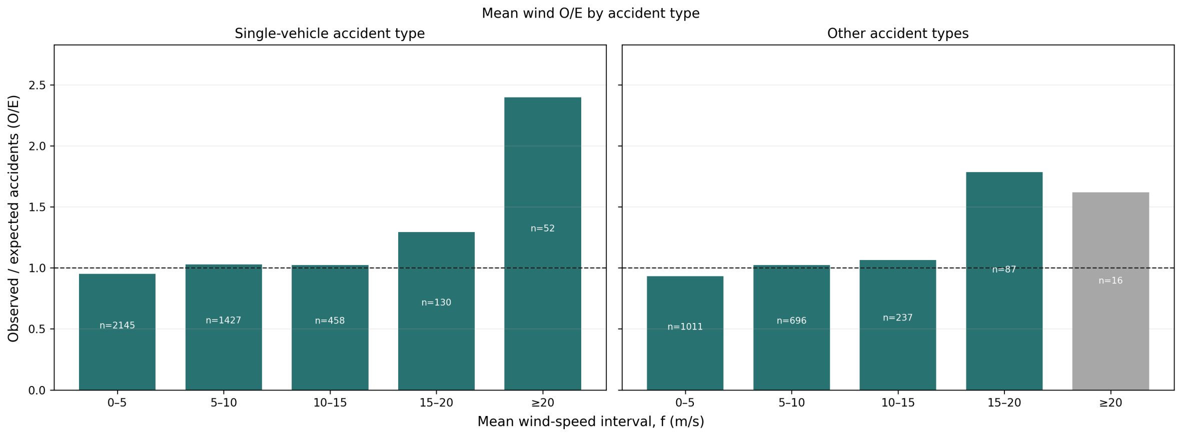

Figure A.6: Mean-wind O/E estimates for single-vehicle run-off-road, rollover, fall, or other events and for all other accident types. At least 20 m/s, the combined O/E is 2.37 for the single-vehicle family and 1.75 for other types. The figure is descriptive and is not a formal test of interaction.

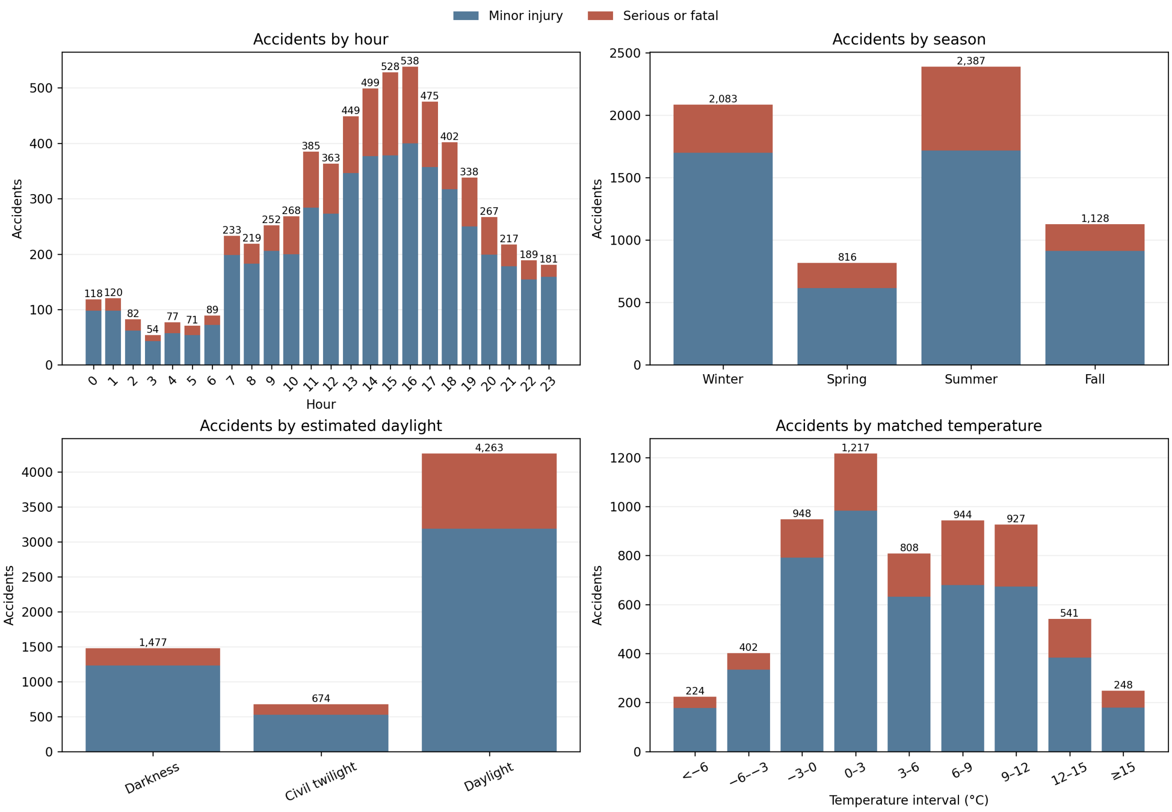

Figure A.7: Raw counts of rural injury accidents by recorded hour, season, estimated daylight, and matched temperature, divided into minor-injury and serious-or-fatal accidents. Labels give total accidents. Hour, season, and daylight use all 6,414 accidents; temperature uses the 6,259 qualifying matches. These panels describe the sample and do not estimate risk.

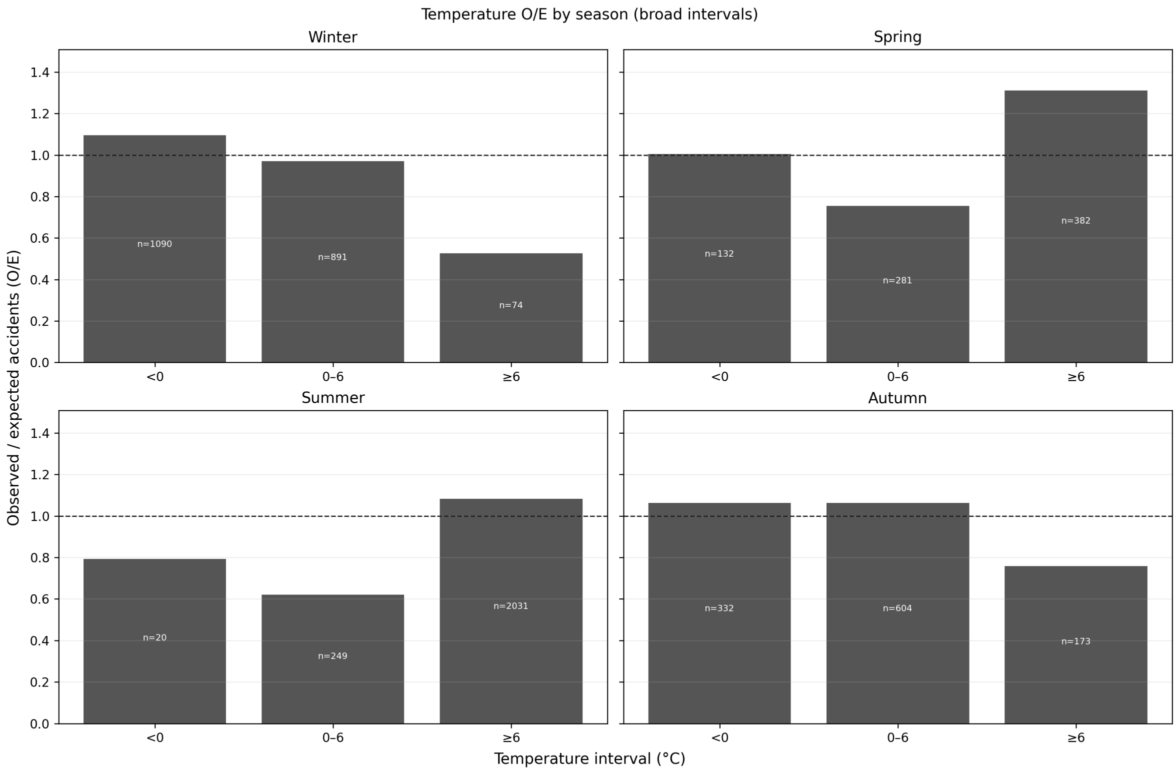

Figure A.8: Temperature O/E by season using three broad intervals formed by summing the fixed temperature bins. The changing direction between seasons shows why the pooled temperature result should be interpreted cautiously. Labels are observed accidents.

Table A.3: Consistency of observed daily PDF traffic with official ADU

Data included	N	Median	P10–P90	$r_{\log}$
All exact matches	2,328	0.975	0.245–1.115	0.942
At least 300 observed days	1,906	0.984	0.240–1.063	0.940
At least 300 days, one counter per section	1,122	1.000	0.925–1.112	0.993

A.6 Full-Day-Mean Daily-Counter Check

Table A.4: Selection of accidents for the observed daily-counter analysis

Selection step	Accidents	Share of 2019–2024 sample
Rural injury accidents in 2019–2024	1,863	100.0%
Exact road-section/year counter candidate	864	46.4%
Assigned counter within 20 km	863	46.3%
Positive count and valid full-day wind	767	41.2%

The main loss occurs because many accident road sections have no daily counter record for the same year. Distance excludes only one additional candidate at 20 km; a further 96 accidents

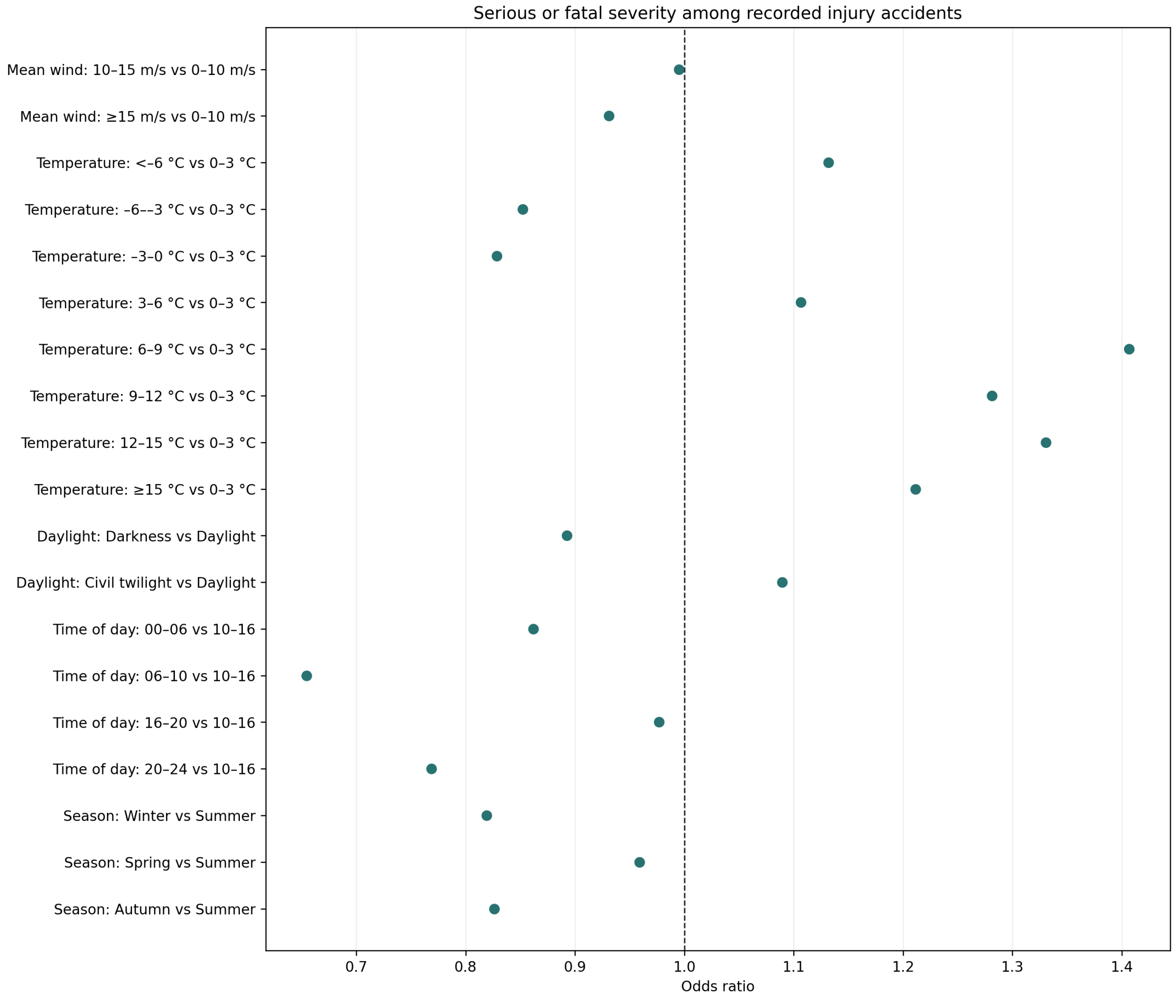

Figure A.9: Adjusted odds of serious-or-fatal rather than minor injury among 6,259 recorded injury accidents with complete wind and temperature matches. The model contains all displayed variables together. It describes severity conditional on an accident and does not estimate accident occurrence.

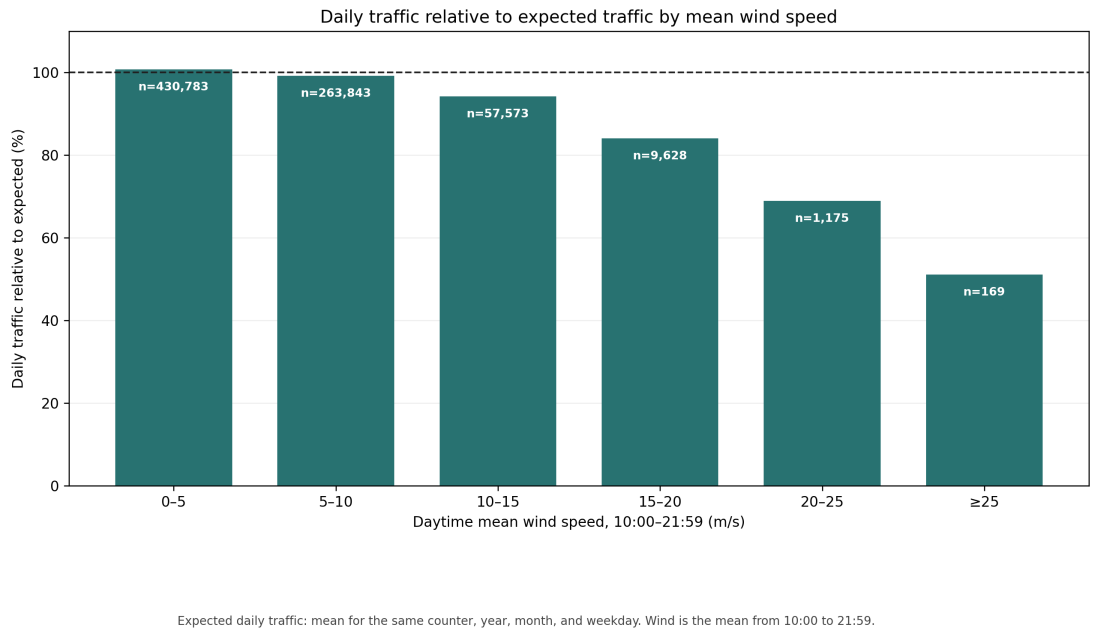

Figure A.10: Vehicles counted per day relative to a typical day at the same counter, year, month, and weekday. Labels show counter-days. Daytime mean wind is calculated from 10:00 to 21:59. The figure describes selected counter sites, not traffic across the national road network.

lack a positive counter count or valid full-day wind on the accident date. No accident is assigned across road sections merely to improve coverage.

Table A.5: Characteristics of accidents retained and excluded by the daily-counter linkage

Characteristic	All	No link	Linked, excluded	Retained
Accidents	1,863	999	102	762
Share of all accidents	100.0%	53.6%	5.5%	40.9%
Distinct road sections	563	374	56	189
Serious or fatal	25.6%	29.4%	18.6%	21.5%
One vehicle	67.7%	75.2%	54.9%	59.7%
Winter	32.6%	28.0%	40.2%	37.5%
Spring	13.0%	11.7%	8.8%	15.4%
Summer	36.0%	41.8%	33.3%	28.6%
Autumn	18.4%	18.4%	17.6%	18.5%

Table A.6: Reasons an exact same-year daily-counter link was unavailable

Reason	Accidents	Share of all accidents
Road section absent from daily-counter files	953	51.2%
Road section countered only in another year	41	2.2%
Historical counter location unavailable	5	0.3%

The selected sample reflects the geography and operating years of the counters; it is not a

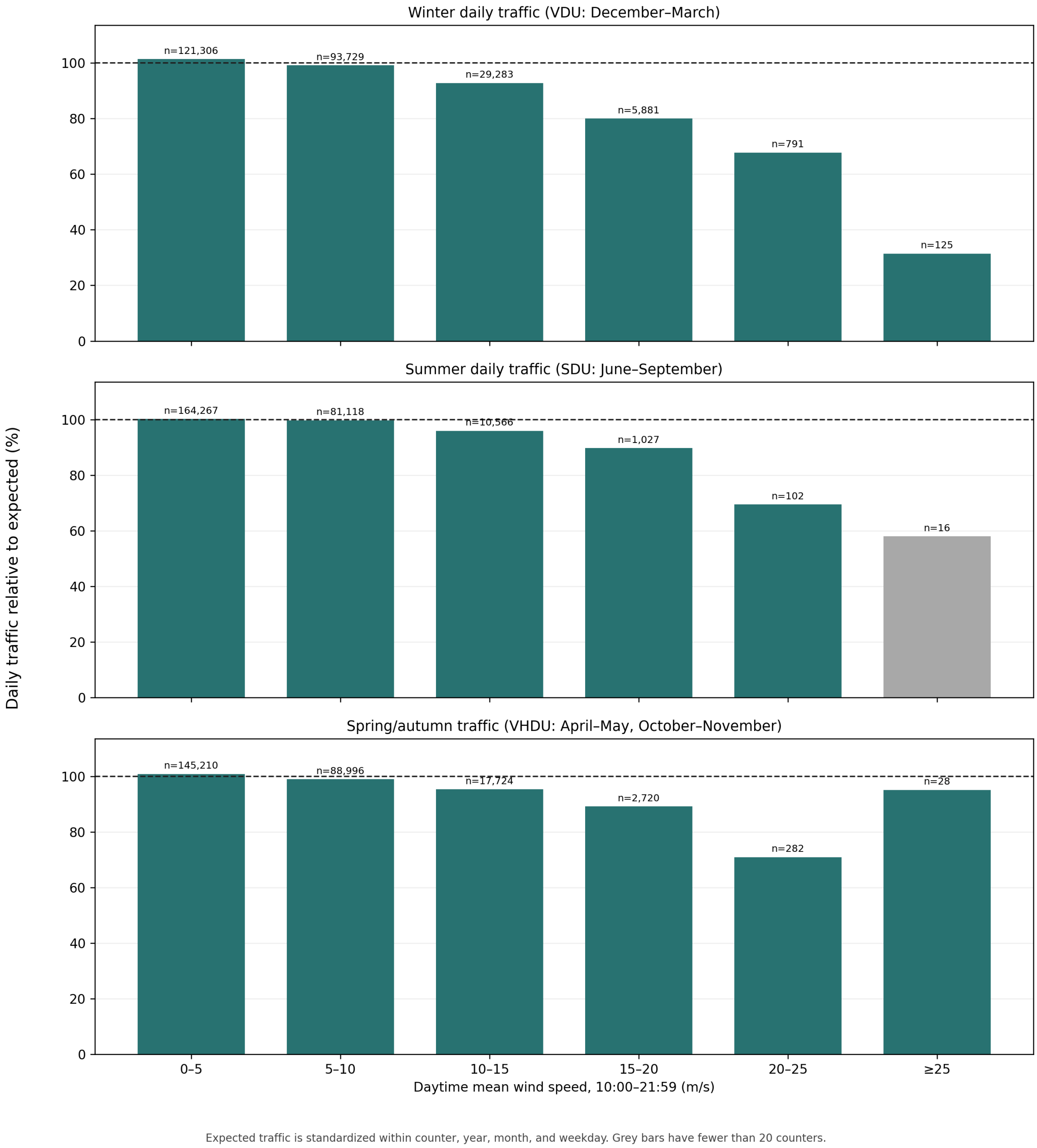

Figure A.11: Daily traffic relative to its calendar expectation by traffic period and daytime mean wind. The decline is present in VDU, SDU, and VHDU, although the highest interval contains only 136, 16, and 29 counter-days respectively. Grey bars contain fewer than 20 counters.

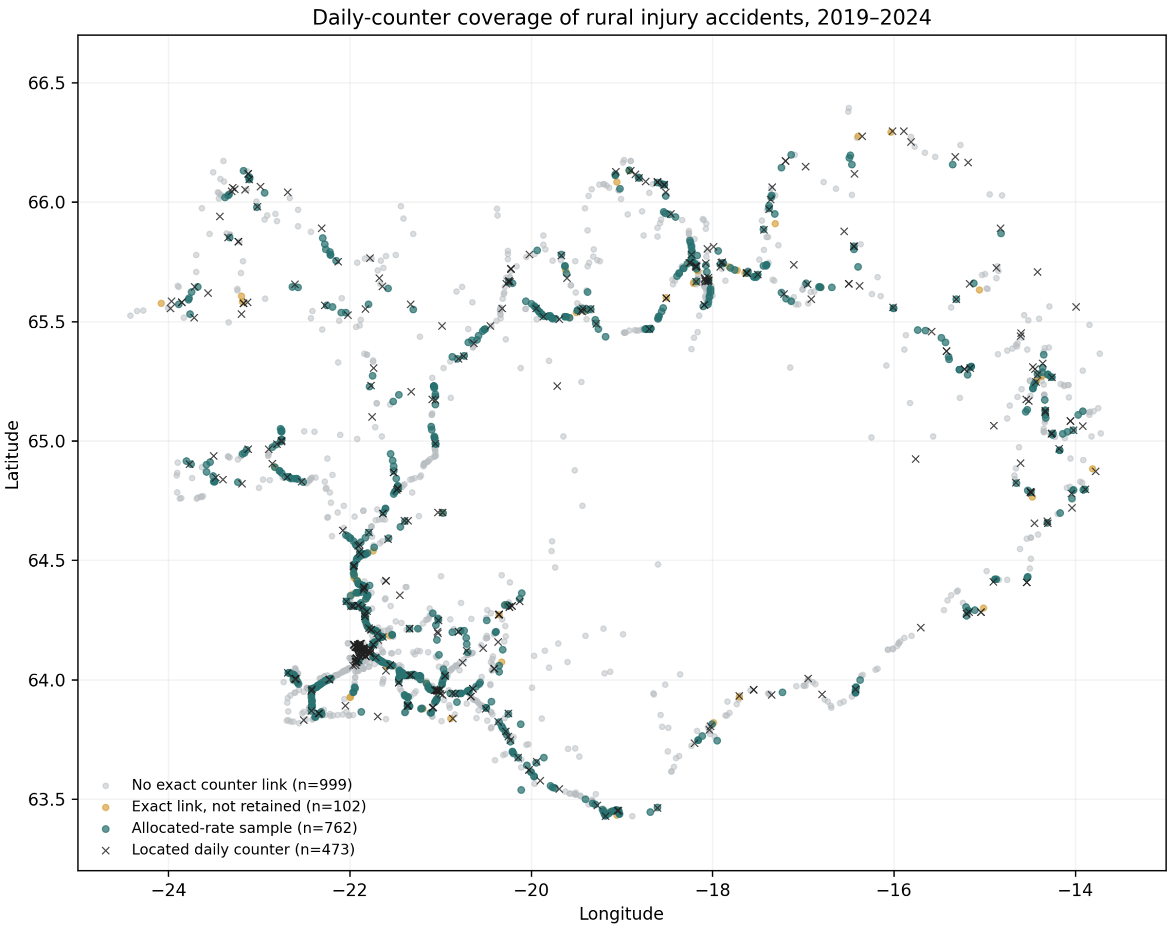

Figure A.12: Locations of the 1,863 rural injury accidents in 2019–2024 and the 473 located daily counters. Colours distinguish the 762 accidents included in the allocated-rate model, the 102 with an exact link but another exclusion, and the 999 without an exact same-year counter link. The map displays the geographic selectivity of the daily-counter sample; it is not a road-network rate map.

representative national sample. It contains fewer summer and serious-or-fatal accidents than the full 2019–2024 rural injury-accident set.

Table A.7: Detailed daily counter-based rural injury-accident rate comparison, 2019–2024

Mean wind	Accidents	Counted vehicles	Counters	Rate ratio (95% CI)
0–5 m/s	397	281,130,848	230	1.00 (reference)
5–10 m/s	271	167,483,620	230	1.06 (0.90–1.24)
10–15 m/s	85	30,314,792	229	1.72 (1.34–2.20)
15–20 m/s	12	2,567,917	169	2.74 (1.52–4.93)
20–25 m/s	1	63,307	61	5.10 (0.70–37.26)
$\geq$25 m/s	1	3,612	7	174.52 (18.78–1,621.80)

The detailed table documents the standard wind intervals but is not used for upper-tail inference: the final two estimates are determined by one accident each and are extremely unstable. The coarse version combines the upper tail and gives a rate ratio of 2.73 (95% CI 1.55–4.78) for at least 15 m/s. Both versions use the same 767 accidents, exact road-section/year assignment within 20 km, and observed 24-hour traffic as the offset. Because a full-day mean does not represent conditions at the accident time, this is only a day-level check.

Table A.8: Upper daily-counter estimate under three assignment distances

Maximum distance	Included accidents	Upper-bin accidents	Rate ratio	95% CI
5 km	585	12	2.91	1.58–5.37
10 km	750	14	2.78	1.58–4.88
20 km	767	14	2.73	1.55–4.78

The estimate remains in the same direction under stricter distance rules. The wide intervals reflect the small number of accidents at full-day mean wind of at least 15 m/s; stability across radii does not remove that limitation.

A.7 Stratified Vehicle-Kilometre Model

Table A.9 gives descriptive absolute rates using estimated vehicle-kilometres from every eligible road section. Table A.10 compares the all-period conditional model with a model restricted to official VDU and SDU traffic periods. The latter excludes April–May and October–November, for which VHDU is derived from ADU, SDU and VDU.

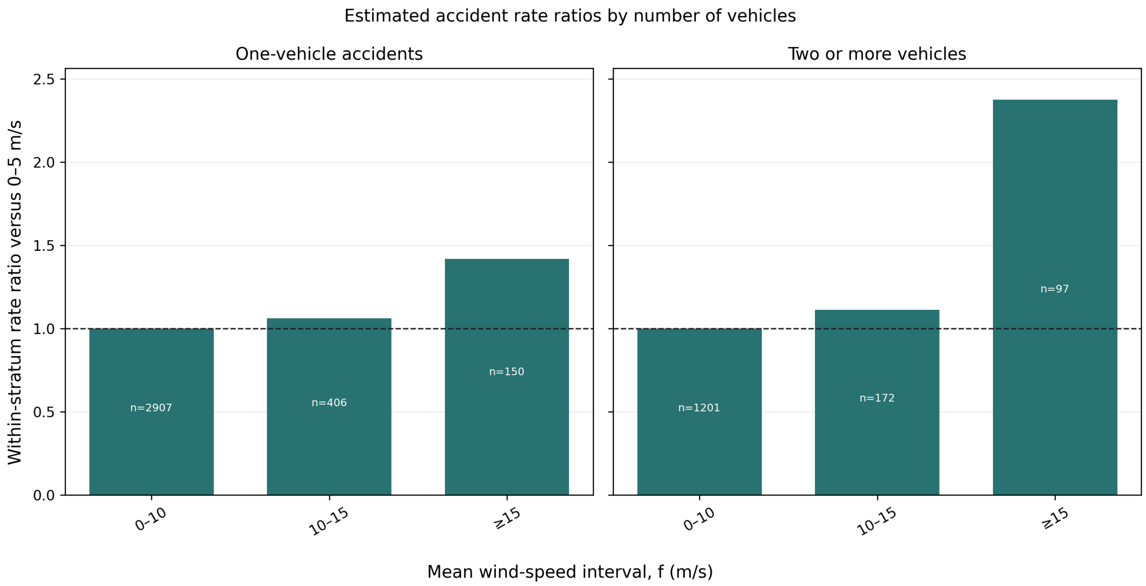

Figure A.13: Estimated mean-wind rate ratios from separate one-vehicle and multiple-vehicle models. The upper category contains 150 and 97 accidents, respectively, in the upper category. Comparisons are within road section, year, and traffic period. Confidence intervals are reported in the result tables.

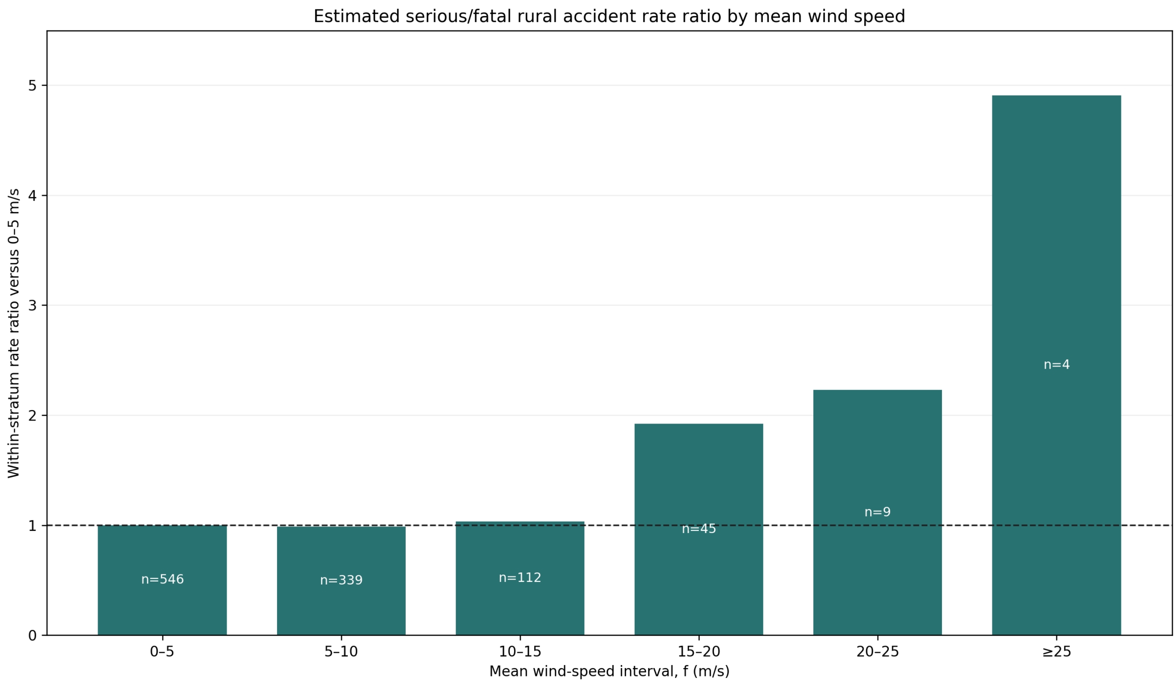

Figure A.14: Estimated serious-or-fatal rural accident rate ratios by mean wind. The model compares wind intervals within the same road section, year, and traffic period. Labels show observed accidents. The four accidents in the highest interval make that estimate imprecise.

Table A.9: Estimated rural injury-accident rate by mean wind speed, all traffic periods

Mean wind	Accidents	Estimated accidents per 100 million vehicle-km
0–5	2,424	8.5
5–10	1,684	13.0
10–15	578	16.6
15–20	184	27.7
20–25	45	40.5
≥25	18	58.6

Table A.10: Conditional rate ratios in the upper mean-wind intervals: all periods versus official VDU and SDU periods only

Mean wind	All periods	Official VDU+SDU only
15–20	1.63 (1.39–1.90), $n = 184$	1.70 (1.41–2.04), $n = 133$
20–25	2.27 (1.68–3.07), $n = 45$	1.82 (1.23–2.70), $n = 26$
≥25	4.95 (3.07–7.98), $n = 18$	2.97 (1.47–6.02), $n = 8$

Table A.11: Comparison and quality checks for the traffic analyses

Check	Data included	Estimate (95% interval)	Records
Rate model, 20-25 m/s	All VDU, SDU, and derived VHDU periods	RR 2.27 (1.68–3.07)	45 accidents
Rate model, 20-25 m/s	Official VDU and SDU periods only	RR 1.82 (1.23–2.70)	26 accidents
Rate model, ≥25 m/s	All VDU, SDU, and derived VHDU periods	RR 4.95 (3.07–7.98)	18 accidents
Rate model, ≥25 m/s	Official VDU and SDU periods only	RR 2.97 (1.47–6.02)	8 accidents
Daily traffic, 20-25 m/s	Retain zero counter-days	68.94% (62.33–74.09)	1,175 days
Daily traffic, 20-25 m/s	Exclude zero counter-days	69.29% (62.70–74.52)	1,090 days
Nonpositive published VDU	Excluded, not imputed	6.57% excluded	1,509 section-years
Nonpositive derived VHDU	Excluded, not imputed	2.40% excluded	552 section-years

Table A.12: Illustrative direction check using observed daily traffic response

Mean wind	Annual-model RR	Daily traffic	Illustrative RR
0–5	1.00	100.7%	1.00
5–10	1.09	99.2%	1.11
10–15	1.13	94.2%	1.21
15–20	1.63	84.0%	1.95
20–25	2.27	68.9%	3.32
≥25	4.95	51.0%	9.77

The last table is a deliberately illustrative calculation. “Daily traffic” is observed traffic as a percentage of the calendar-standardised expectation on days classified by full-day mean wind.

The illustrative rate ratio multiplies the annual-model rate ratio by the 0–5 m/s percentage divided by the percentage for the displayed interval. Because the daily data cover selected counters in 2019–2024 and do not contain traffic by ten-minute interval, these values are not corrected estimates or additional model results.

The two models agree that the rate ratio is higher from 15 m/s. The upper intervals differ because the official-only analysis excludes VHDU and has few accidents. It shows how the estimated rate depends on the estimated rate on the traffic-period assumption.

The four-season rate model is reported in the main results using wider wind intervals to avoid very small cells. A vehicle-group rate model is not reported because the upper intervals become too sparse for a balanced analysis. The descriptive O/E vehicle-group figure above shows the limited evidence available.